

A Casimir–Secular Electroweak Mass Ratio and Its String/Kaluza–Klein Interpretation Problem

Alejandro Rivero¹

¹*EUPT, Universidad de Zaragoza*

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Abstract

We formulate the Rivero–de Vries electroweak observation as a two-branch secular problem built from a Casimir variable $J = s(s + 1)$. The positive root of $x^2 + Jx - J = 0$, evaluated at $J = 3/4$ and $J = 2$, gives $1 - x_+(3/4)/x_+(2) = 0.22310132\dots$, close to the weak angle defined by the W/Z pole-mass ratio. The same secular equation has a negative branch whose natural vector-spectrum normalization gives scales near the Higgs/order-parameter side of electroweak symmetry breaking. The paper treats these facts as a conceptual clue about where a string/Kaluza–Klein spectral datum might attach to the Standard Model. The proposed attachment point is the low-energy pole spectrum. The comparison point is the grand-unified value $\sin^2 \theta_W = 3/8$, a high-scale group-theoretic datum subsequently related to experiment by renormalization-group flow. The manuscript returns to the early dual-model and Kaluza–Klein ambition that string theory should organize observed particle structure, and keeps the modern distinctions among pole masses, running parameters, compactification spectra, and symmetry-breaking data. We present the algebraic construction, the electroweak projective interpretation in which the Higgs vacuum scale is radial, and the analytical tasks required for a dynamical derivation. The central open calculation is the derivation of the ordered assignment $(J_W, J_Z) = (3/4, 2)$ from the gauge-Higgs/order-parameter sector or from a compactification boundary problem.

I. INTRODUCTION

The electroweak sector contains a dimensionless pole-spectrum ratio,

$$s_{\text{pole}}^2 = 1 - \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2}, \quad (1)$$

which is fixed experimentally by the pole positions of the charged and neutral weak vector bosons. In the Standard Model at tree level the same ratio is written as an angular datum in coupling space, while the Higgs vacuum expectation value fixes the common radial scale of the vector masses. This separation between projective and radial data gives the working language of the present paper.

The Rivero–de Vries observation starts from a mass operator built from Poincare Casimir data and produces a dimensionless two-branch equation. In the normalized notation used below, the equation is

$$x^2 + Jx - J = 0, \quad J = s(s + 1). \quad (2)$$

The positive branch gives

$$s_{\text{dV}}^2 = 1 - \frac{x_+(3/4)}{x_+(2)} = 0.22310132\dots \quad (3)$$

The original note emphasized the proximity of this number to the mass-shell weak angle and motivated the operator through Casimir invariants and high-spin behavior [1]. The present manuscript treats the numerical proximity as descriptive. It records a structural clue. Eq. (3) has clue status in this draft. The central question asks which physical object carries the square root in Eq. (2), and why that object is read at the low-energy W/Z pole ratio.

This placement question is the conceptual core. A standard grand-unified lineage uses representation theory to obtain a clean weak-angle value at a high scale. In $SU(5)$, and in $Spin(10)$ with the usual fermion content, the famous group-theoretic value is $\sin^2 \theta_W = 3/8$; contact with the observed low-energy angle is then made through running and thresholds [20]. The Rivero–de Vries construction reverses the burden. Its clean value is close to the pole-mass ratio itself. Therefore the manuscript must explain why a spectral datum should attach to the pole spectrum. A unification-scale boundary condition would define a different reading of the same algebra.

The string-theory posture uses the older dual-model and Regge ambition as part of the physical context. Open strings, endpoint data, Kaluza–Klein spectra, and boundary conditions are treated as possible mechanisms for organizing low-energy particle structure. String theory also incorporates Kaluza–Klein ideas, grand-unified representations, and supersymmetric structure in transmuted form [3]. The working problem is to ask whether those tools can produce an electroweak secular datum at the observed pole spectrum. The numerical clue remains a clue until that mechanism is written.

The construction has four layers. The first layer is algebraic: a two-channel secular equation has roots whose ratio is fixed by the Casimir variable. The second layer is electroweak: the W/Z mass ratio is a projective datum along the broken-to-unbroken ray of the gauge-Higgs system. The third layer is spectral: the square root must be assigned to a pole, running, compactification, or boundary-condition object. The fourth layer is dynamical: strings, branes, Kaluza–Klein boundary conditions, or singular G_2 geometries may generate the secular equation or its branch structure.

The status of the argument is deliberately stratified. The algebraic calculation is explicit.

The pole-mass comparison is a descriptive clue whose source audit remains a separate task. The assignment of $J = 3/4$ to the charged W comparison and $J = 2$ to the neutral Z comparison is the central open analytical calculation unless it is derived in Sec. IV. The negative branch is retained because it may encode the scalar/order-parameter side of electroweak symmetry breaking.

The paper is organized as follows. Section II gives the secular operator and its roots. Section III defines the pole observable and the placement problem. Section IV formulates the projective electroweak interpretation. Section V records the negative branch and its possible Higgs-scale role. Sections VI–XIII develop the string, endpoint, Kaluza–Klein, and G_2 mechanism tests in a common two-channel language. Sections XIV and XV delimit the flavor and global-gauge-group issues. Section XVI gives the status of the analytical program and its conceptual tests.

Section status. This introduction states the conceptual placement problem and labels the central unproved step.

II. CASIMIR-SECULAR MASS OPERATOR

Let $J = s(s + 1)$ be the spin-Casimir eigenvalue. The normalized two-channel operator used in this manuscript is

$$\frac{Q(J)}{\mu^2} = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}. \quad (4)$$

Its characteristic equation is

$$\det \left(xI - \frac{Q(J)}{\mu^2} \right) = x^2 + Jx - J = 0. \quad (5)$$

The two roots are

$$x_{\pm}(J) = \frac{-J \pm \sqrt{J^2 + 4J}}{2}. \quad (6)$$

They obey

$$x_+(J) + x_-(J) = -J, \quad x_+(J)x_-(J) = -J. \quad (7)$$

For $J > 0$, the positive branch is positive and the negative branch is negative.

Historically, the electroweak number is obtained by sampling the positive branch at the two Casimir values

$$J_H = \frac{3}{4}, \quad J_{\text{adj}} = 2. \quad (8)$$

This gives

$$s_{\text{dV}}^2 = 1 - \frac{x_+(3/4)}{x_+(2)}. \quad (9)$$

The expression is dimensionless; the overall scale μ cancels. The cancellation is essential for an electroweak interpretation because Eq. (9) can then be compared with a projective W/Z mass ratio. In the electroweak reading developed here, $J_H = 3/4$ is the Higgs/order-parameter doublet invariant and $J_{\text{adj}} = 2$ is the adjoint/current invariant. The ordered use of these two samples in the W/Z quotient remains the central derivation problem.

The relation to the original Rivero operator is obtained by restoring the Poincare Casimir eigenvalues $C_1 = m^2$ and $C_2 = -m^2 s(s+1)$. In the original construction, a mass-squared operator is formed from combinations of Casimir invariants, with a high-spin condition designed to preserve the asymptotic mass behavior relevant for Regge trajectories [1]. Equation (4) is the reduced dimensionless representative of that branch problem.

A. What the algebra fixes

The operator in Eq. (4) fixes three algebraic facts. First, the trace is $-J$. Second, the determinant is $-J$. Third, the two eigenvalues are tied by the same invariant:

$$\text{tr} \frac{Q(J)}{\mu^2} = -J, \quad \det \frac{Q(J)}{\mu^2} = -J. \quad (10)$$

Thus a physical derivation has to explain a simultaneous trace and determinant assignment. A model that gives only a square-root branch, or only a J -dependent diagonal term, leaves the DeVries construction incomplete. The required two-channel matrix has the schematic form

$$\mathcal{M}_J = \begin{pmatrix} 0 & \kappa_J \\ \kappa_J & -\tau_J \end{pmatrix}, \quad \kappa_J^2 = J, \quad \tau_J = J. \quad (11)$$

This is the compact target for the endpoint, Kaluza–Klein, and G_2 sections.

The square-root radical is then the discriminant of this two-channel problem:

$$\Delta_J = J^2 + 4J. \quad (12)$$

The radical has to attach to a physical object. The radical-placement section lists the main options. The current manuscript follows the pole-spectrum placement because the positive and negative branches are interpreted as a single low-energy spectral clue.

B. Minimal assumptions and theorem owed

A complete derivation should state the minimal assumptions that select Eq. (4). The current minimal list is:

1. dependence on available Poincare or internal invariants;
2. dimensional consistency after restoring the mass scale μ ;
3. a two-branch secular structure;
4. preservation of the high-spin or Regge asymptotic datum in the originating problem;
5. a trace/determinant condition equivalent to Eq. (10).

These assumptions define a research program; the uniqueness theorem remains open. The theorem to prove would read: within a stated class of two-channel operators built from the allowed invariants, the high-spin condition and trace/determinant constraints select Eq. (4) up to similarity transformations and normalization.

This theorem would also clarify the status of the negative branch. Since $x_-(J)$ is fixed by the same trace and determinant as $x_+(J)$, a derivation of Eq. (4) automatically derives both branches. The electroweak interpretation can then ask which fields or boundary data realize the two eigenvectors.

Section status. The eigenvalue algebra is derived. The uniqueness of the operator under the stated constraints is an open analytical subtask.

III. POLE OBSERVABLE AND ELECTROWEAK SCHEME SEPARATION

The observable used for comparison is the pole-mass ratio

$$s_{\text{pole}}^2 = 1 - \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2}. \quad (13)$$

This ratio is distinct from a running $\overline{\text{MS}}$ weak angle and from effective weak mixing angles extracted from neutral-current asymmetries. The manuscript therefore treats Eq. (13) as the primary target and later relates other schemes to it.

The following placement convention will be used. The Rivero–de Vries radical is read as a spectral object attached to the W/Z pole data. Other readings could attach the same radical

to a high-scale boundary value, to a running $\overline{\text{MS}}$ coupling, or to an internal compactification eigenvalue. Each reading changes the physical content of the proposal. The pole reading gives the word “spectrum” its literal electroweak meaning.

A vector-boson pole is defined by the complex pole of the propagator,

$$s_V = M_{V,\text{pole}}^2 - iM_{V,\text{pole}}\Gamma_{V,\text{pole}}. \quad (14)$$

Equivalently, s_V is the zero of the inverse transverse propagator after renormalization,

$$\Delta_{T,V}^{-1}(s_V) = 0. \quad (15)$$

When experimental inputs are quoted in a variable-width Breit–Wigner convention, the conversion to pole parameters must be performed before using Eq. (13). Martin and Robertson summarize the common relation for the vector bosons as

$$M_{V,\text{BW}}^2 = M_{V,\text{pole}}^2 + \Gamma_{V,\text{pole}}^2, \quad V = W, Z. \quad (16)$$

This is one of the source-audit tasks of the project [10].

A. Scheme ledger

The project uses the following ledger before comparing any weak-angle quantity:

Object	Definition level	Role in this manuscript
s_{pole}^2	complex-pole mass ratio	primary low-energy spectral target
Breit–Wigner masses	experimental line-shape convention	input convention requiring Eq. (16)
$\hat{s}^2(Q)_{\overline{\text{MS}}}$	running coupling ratio in a renormalization scheme	scheme translation and loop-scale comparison
$\sin^2 \theta_{\text{eff}}^\ell$	neutral-current effective angle	phenomenological comparison after electroweak corrections
GUT boundary value	high-scale representation datum	comparison class for structural weak-angle values
compactification eigenvalue	internal spectral datum	possible origin for $\mathcal{O}_J$ after a dynamical derivation

The pole reading selects the first row. The remaining rows are translation targets. A future source-audited electroweak comparison should begin from the quoted experimental convention, map to complex pole quantities, form Eq. (13), and then compare with Eq. (9).

The project seed comparison uses

$$M_W = M_Z \sqrt{1 - s_{\text{dV}}^2}. \quad (17)$$

For the seed value $M_Z = 91.1880$ GeV, this gives $M_W = 80.3748$ GeV. These numbers are placeholders inherited from the working draft and must be audited against the current electroweak source set before final submission. The descriptive comparison places the construction near conventional pole-spectrum averages and away from the CDF-II W-mass value [11]. The comparison has clue status in this draft. The numerical appendix records the inherited seed arithmetic for provenance.

The Standard Model parameter literature distinguishes several vacuum and renormalization conventions. In particular, pure $\overline{\text{MS}}$ treatments may define the Higgs vacuum expectation value as the minimum of the loop-corrected effective potential in Landau gauge, eliminating tadpole diagrams at the cost of gauge-choice dependence in intermediate parameters [10]. Such scheme details are relevant for comparing a mass-ratio statement with Lagrangian parameters, while Eq. (13) remains the pole-spectrum target.

Loop effects are therefore a conceptual discriminator. Corrections of order α/π are of the same kind of size one expects when moving between pole, effective, and running definitions. They ask what object the radical belongs to and at what scale it is to be read. A high-scale reading would make the DeVries value analogous to a GUT boundary datum. The pole-data reading makes it a low-energy spectral datum and transfers the burden to string/Kaluza–Klein dynamics.

B. Referee-facing claim

The formal claim of this section is limited:

$$s_{\text{dV}}^2 \quad \text{is compared first with} \quad 1 - \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2}. \quad (18)$$

Every comparison to $\overline{\text{MS}}$ inputs, effective angles, or line-shape masses requires an explicit map. Thus a disagreement among schemes diagnoses the placement of $\sqrt{J^2 + 4J}$ and the scale at which the radical is read.

Section status. The observable and placement convention are defined. The numerical input table requires a current electroweak source audit before final claims.

IV. ELECTROWEAK EMBEDDING AS A PROJECTIVE DATUM

The gauge-Higgs sector gives the tree-level vector masses from the Higgs kinetic term

$$(D_\mu H)^\dagger (D^\mu H), \quad D_\mu = \partial_\mu - \frac{ig}{2} \sigma^a W_\mu^a - \frac{ig'}{2} B_\mu, \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h \end{pmatrix}. \quad (19)$$

The charged fields

$$W_\mu^\pm = \frac{1}{\sqrt{2}} (W_\mu^1 \mp i W_\mu^2) \quad (20)$$

receive a mass term $g^2 v^2 W_\mu^+ W^{-\mu} / 4$. The neutral fields have mass matrix

$$\frac{v^2}{8} \begin{pmatrix} W_\mu^3 & B_\mu \end{pmatrix} \begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix} \begin{pmatrix} W^{3\mu} \\ B^\mu \end{pmatrix}. \quad (21)$$

It is diagonalized by

$$Z_\mu = \cos \theta_W W_\mu^3 - \sin \theta_W B_\mu, \quad A_\mu = \sin \theta_W W_\mu^3 + \cos \theta_W B_\mu, \quad \tan \theta_W = \frac{g'}{g}. \quad (22)$$

This gives

$$M_W^2 = \frac{g^2 v^2}{4}, \quad M_Z^2 = \frac{(g^2 + g'^2) v^2}{4}, \quad M_\gamma^2 = 0. \quad (23)$$

The ratio is

$$1 - \frac{M_W^2}{M_Z^2} = \frac{g'^2}{g^2 + g'^2}. \quad (24)$$

Thus the vacuum scale v fixes the radial size of the massive vector pair, while the quotient fixes a projective direction in the two-coupling plane [5, 9].

A. Custodial structure and the electroweak ray

The Standard Model doublet can be packaged as the bidoublet

$$\Phi_H = \begin{pmatrix} H^{0*} & H^+ \\ -H^{+*} & H^0 \end{pmatrix}. \quad (25)$$

In the scalar sector this makes the approximate global $SU(2)_L \times SU(2)_R$ structure visible; the vacuum proportional to the identity preserves the diagonal custodial $SU(2)$ subgroup

[9]. Hypercharge gauging and Yukawa splittings break the symmetry, while the tree-level doublet mass matrix preserves the relation

$$\rho = \frac{M_W^2}{M_Z^2 \cos^2 \theta_W} = 1. \quad (26)$$

This is the electroweak reason the DeVries comparison should be a projective statement about the W/Z pair. The broken-to-unbroken deformation scales the vector masses with v while preserving the coupling-space direction.

The DeVries proposal is read in the same projective language:

$$\frac{M_W^2}{M_Z^2} \longleftrightarrow \frac{x_+(3/4)}{x_+(2)}. \quad (27)$$

The common scale is then supplied by the electroweak vacuum or, in a deeper model, by the compactification or brane scale that generates the effective gauge-Higgs sector.

The unbroken limit is the ray

$$v \rightarrow 0 \quad \text{with} \quad \frac{g'^2}{g^2 + g'^2} \text{ fixed.} \quad (28)$$

Along this ray, M_W and M_Z vanish together and the photon remains massless. This is the electroweak deformation relevant to the construction. The projective datum remains meaningful as the limiting direction even when the radial mass scale is taken to zero.

The central analytical question is the ordered assignment

$$(J_W, J_Z) = \left(\frac{3}{4}, 2 \right). \quad (29)$$

The electroweak gauge fields are gauge-algebra fields before symmetry breaking: W^1, W^2, W^3 live in the adjoint of $SU(2)_L$, while B is the gauge field of the abelian $U(1)_Y$ factor. The value $J = 2$ has a direct $SU(2)_L$ adjoint reading through $T(T+1)$ with $T = 1$. The value $J = 3/4$ belongs to the Higgs/order-parameter side, where the symmetry-breaking field is an $SU(2)_L$ doublet with $T = 1/2$ and $T(T+1) = 3/4$.

This yields a candidate route:

$$J_Z = T_{\text{adj}}(T_{\text{adj}} + 1) = 2, \quad J_H = T_H(T_H + 1) = \frac{3}{4}. \quad (30)$$

The remaining calculation is the ordered sampling rule that turns the pair (J_H, J_{adj}) into the W/Z quotient in Eq. (27). A completed derivation must start from the gauge-Higgs Lagrangian, a current-algebra construction, or a compactification boundary problem and produce Eq. (29) from the same determinant that supplies the negative branch.

B. The assignment problem as a theorem target

The theorem target can be stated mechanism-independently. Let $\mathcal{E}$ be the electroweak gauge-Higgs system, including its projective W/Z pole data and scalar order-parameter channel. A DeVries derivation is a map

$$\mathcal{D} : \mathcal{E} \longrightarrow \mathcal{M}_J = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix} \quad (31)$$

with the following outputs:

$$\mathcal{D}(\text{order-parameter channel}) \mapsto J_H = 3/4, \quad (32)$$

$$\mathcal{D}(\text{adjoint/current channel}) \mapsto J_{\text{adj}} = 2, \quad (33)$$

$$\frac{M_W^2}{M_Z^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})}. \quad (34)$$

The theorem must also account for the negative branch in the same basis. This prevents the vector ratio from being fitted by one argument and the scalar clue by another.

A natural current to test is the Higgs isospin current

$$\mathcal{J}_\mu^a = iH^\dagger \frac{\sigma^a}{2} D_\mu H - i(D_\mu H)^\dagger \frac{\sigma^a}{2} H, \quad (35)$$

which transforms as an adjoint object while being built from the doublet order parameter. This current has the right qualitative structure for the ordered pair: doublet data enter the source, adjoint data enter the current. Turning this observation into Eq. (34) is the electroweak calculation owed.

Section status. The projective interpretation follows from the electroweak mass matrix. The adjoint and doublet Casimirs are identified. The ordered sampling rule in Eq. (29) remains the primary open calculation.

V. NEGATIVE BRANCH AND THE SCALAR SIDE OF THE PROBLEM

The negative branch in Eq. (6) is fixed by the same secular equation as the positive branch. It is therefore part of the clue. Hans de Vries identified the positive-branch proximity to the mass-shell weak angle. The same two-root object also contains negative eigenvalues near the scalar/order-parameter scales of electroweak symmetry breaking when the positive branch

is normalized by the vector spectrum. That observation is the main reason the numerics deserve conceptual attention.

The root identities imply

$$x_-(J) = -J - x_+(J) = -\frac{J}{x_+(J)}. \quad (36)$$

Any physical use of the positive branch automatically carries a partner branch.

The Higgs potential can be written as

$$V(H) = m^2 H^\dagger H + \lambda (H^\dagger H)^2, \quad m^2 < 0 \quad (37)$$

in the broken phase. The sign of the quadratic term makes the negative branch tempting as an order-parameter datum. If the normalization is fixed by $M_Z^2 = \mu^2 x_+(2)$, the two negative magnitudes are

$$M_{-,1} = M_Z \sqrt{\frac{|x_-(2)|}{x_+(2)}}, \quad M_{-,1/2} = M_Z \sqrt{\frac{|x_-(3/4)|}{x_+(2)}}. \quad (38)$$

The first lies near the electroweak vacuum scale convention $v/\sqrt{2}$, and the second lies near the Higgs pole-mass range in the seed arithmetic of the project. They are unresolved branch-scale identifications. The secular equation points at the vector pole ratio and at the scalar side of electroweak symmetry breaking at the same time.

A valid identification must be gauge-invariant and scheme-controlled. Possible targets include m^2 , λv^2 , the Higgs pole mass, or an eigenvalue in a compactification problem. The branch values associated with $J = 3/4$ and $J = 2$ generate definite dimensionless numbers. The map from those numbers to scalar observables remains open.

A. Logical role of the negative root

The negative root changes the character of the clue. The positive branch by itself would be a ratio of two selected positive eigenvalues. The full quadratic gives a paired object: a pole-ratio candidate and a scalar-side candidate arise from one secular equation. This pairing is the conceptual reason to continue the project. A dynamical theory that produces Eq. (2) has to say what both branches mean in the same basis.

The two-root structure can be stated as a spectral ledger,

$$\mathcal{M}_J \longrightarrow \begin{cases} x_+(J) & \text{projective vector datum,} \\ x_-(J) & \text{scalar/order-parameter datum.} \end{cases} \quad (39)$$

The first line is the pole-ratio reading developed in Sec. III. The second line is a target for the Higgs, boundary, brane, or compactification sector. The ledger is meaningful because both entries come from the same diagonalization. Any proposed derivation that keeps the positive branch acquires the negative-branch obligation automatically.

This obligation also sharpens the electroweak assignment. The ordered pair $(J_H, J_{\text{adj}}) = (3/4, 2)$ already suggests two electroweak structures: the doublet order parameter and the adjoint current. The negative branch asks whether the doublet/order-parameter side can be read as a scalar instability, a vacuum scale, or a deformation mode in the same operator that gives the vector pole quotient. The scalar reading therefore ties together three open calculations: the ordered J -assignment, the determinant derivation, and the branch-to-Higgs map.

B. Gauge-invariant scalar language

The scalar sector has several layers. The Higgs doublet H itself is gauge-covariant. The radial composite

$$\varphi^2 = 2H^\dagger H \quad (40)$$

is the natural gauge-invariant local order-parameter variable for this manuscript's purpose. In the broken phase, a gauge choice may represent the vacuum by a real radial expectation value, yet the physical statement should be expressible through invariant quantities, pole positions, or matched parameters. The negative branch therefore requires a defined scalar functional before it can carry a physical claim.

A useful abstract target is

$$\mathcal{F}_{\text{sc}} = \mathcal{F}_{\text{sc}}(H^\dagger H, (D_\mu H)^\dagger D^\mu H, V(H), \text{boundary or compactification data}), \quad (41)$$

with a branch assignment

$$\mathcal{F}_{\text{sc}}(J_\star) = C_{\text{sc}} |x_-(J_\star)|. \quad (42)$$

Here $J_\star$, C_{sc} , and the scheme are part of the theorem. The expression may describe a pole mass, a curvature of the scalar potential, a vacuum-ray coordinate, a brane separation, or a compactification deformation. The manuscript should treat all of these as candidate targets until one route derives a specific object.

The scalar pole target is the most restrictive because it demands the same pole discipline as the vector comparison. The potential target is more sensitive to matching conventions, because m^2 , λ , and v depend on the renormalization scheme and on how the effective potential is defined. The compactification target is broader: $x_-(J)$ may first appear as an eigenvalue or deformation parameter in a higher-dimensional problem, with the four-dimensional Higgs interpretation emerging after reduction.

C. Vacuum ray and branch pairing

The electroweak ray separates scale from direction. Along the broken-to-unbroken ray, the radial coordinate scales the vector masses, while the ratio

$$\frac{M_W^2}{M_Z^2} = \frac{g^2}{g^2 + g'^2} \quad (43)$$

is projective at tree level. The DeVries quotient addresses this projective datum. The negative branch is naturally assigned to the scalar side because it shares the same two-channel operator while carrying the opposite eigenvalue.

In a route with fields h_J and a_J , the branch vectors have the schematic form

$$u_{\pm}(J) = \alpha_{\pm}(J)h_J + \beta_{\pm}(J)a_J. \quad (44)$$

The scalar-side reading becomes a statement about the h_J component of $u_-(J)$. The vector-side reading becomes a statement about the projective pole ratio extracted from $u_+(J)$. A complete derivation should therefore give a branch basis, an inner product, and a normalization. These data determine whether the negative eigenvector genuinely selects an order-parameter amplitude or whether it belongs to an auxiliary sector of the chosen source theory.

The electroweak ray also supplies a useful consistency check. Taking the radial vacuum coordinate toward the symmetric point should collapse the vector masses together with the Higgs order parameter. The DeVries ratio can remain as a projective direction in coupling or spectral space. A scalar interpretation of x_- should respect this ray structure by scaling with the radial data or by controlling the deformation that sets the radial scale.

D. Candidate scalar targets

The Standard Model scalar sector gives several inequivalent objects that can carry a scalar-side interpretation [8, 9]. They have different scheme and gauge properties:

Target	Meaning	DeVries requirement
m^2	quadratic term in the Higgs potential	define the scheme and gauge-invariant extraction
λv^2	curvature scale of the scalar potential along the radial mode	connect branch value to the electroweak ray
$M_{h,\text{pole}}^2$	scalar pole position	derive the scalar pole from the same determinant as W/Z
radion/modulus eigenvalue	compactification scalar or boundary deformation	identify the compact operator and its normalization
brane separation mode	endpoint or brane order parameter	show coupling to the adjoint/current channel

The pole interpretation is the sharpest target. It asks the same object that governs the vector pole quotient to generate a scalar pole or scalar instability. The potential interpretation is more flexible, because m^2 , λ , and v depend on conventions and matching. The compactification interpretation is broader still; it may identify the negative branch with a modulus or boundary deformation whose four-dimensional scalar meaning emerges after reduction.

E. Route-specific scalar maps

The endpoint route maps the negative branch to brane scalar data. In a D-brane system, transverse positions of branes appear as scalar fields on the worldvolume, while strings stretched between branes acquire masses controlled by separation data [2, 3]. A DeVries endpoint derivation would identify $u_-(J)$ with a brane-Higgsing direction, a stretched-string scalar, or a boundary order-parameter field. The map has to use the same endpoint algebra that supplies the current trace and off-diagonal product.

The interval route maps the negative branch to boundary or modulus data. A five-dimensional gauge field supplies four-dimensional vector modes and scalar components; the compactification radius, brane terms, and boundary conditions determine the light spectrum [7, 12]. A DeVries interval derivation would identify $u_-(J)$ with a boundary scalar amplitude, an A_5 -like scalar, a radion-like deformation, or a brane-localized order parameter. The map has to preserve the same boundary determinant that gives the W/Z branch quotient.

The G_2 route maps the negative branch to singularity deformation data. Local G_2 constructions place gauge sectors and charged matter at singular loci, and Higgs-bundle descriptions encode matter localization and interaction data through functions on an associative three-cycle [14–16]. A DeVries G_2 derivation would identify $u_-(J)$ with a deformation parameter, a localized modulus, a critical value of f_Q , or a charge-density functional. The same local geometry then has to produce the adjoint/current and order-parameter entries of the kernel.

These maps give the negative branch a practical role in route selection. A candidate route that produces the positive-branch quotient and leaves the negative eigenvector with a scalar or deformation interpretation becomes stronger. A route that produces a positive branch while assigning the negative branch to an unrelated auxiliary variable still teaches the manuscript where the physical claim should stop.

F. Eigenvectors and branch meaning

The two roots also have eigenvectors. For the matrix in Eq. (4), an eigenvector with eigenvalue x can be represented by

$$v_x \propto \begin{pmatrix} \sqrt{J} \\ x \end{pmatrix}. \quad (45)$$

Thus the branch assignment is a statement about channel composition. The positive and negative roots are mixtures of the same two basis amplitudes. A scalar interpretation of x_- should therefore identify the basis in which one component is the vector/current side and the other is the order-parameter or boundary side.

A useful criterion is the following. If the same two-channel determinant produces both the vector pole ratio and the scalar instability, then the determinant should appear in an

effective action, boundary condition, or self-energy equation. The next calculation is to write such a determinant explicitly and identify which field or mode occupies the second channel.

Section status. The negative branch algebra is exact. Its Higgs interpretation is conjectural and requires a dynamical map.

VI. STRING, KALUZA–KLEIN, AND G_2 COMPLETION PROBLEM

The secular equation has the form of a low-level branch splitting that preserves a high-spin datum. This is the reason string and Regge language enter the project. The original Rivero construction imposed a high-spin condition motivated by Regge behavior [1]. A string interpretation must therefore preserve the asymptotic slope while explaining why the low levels split through the two-channel determinant.

The string-theory input is concrete. Quantized strings generate towers, level matching, endpoint sectors, winding, and compactification charges. A compact circle already contains the basic Kaluza–Klein mechanism: a higher-dimensional metric reduces to a lower-dimensional metric, a gauge field, and a radius modulus; Fourier modes along the circle acquire masses n^2/R^2 and charges under the gauge field descending from the metric [2]. The string worldsheet adds winding sectors, and winding modes couple to the vector descending from the antisymmetric tensor. This is the first useful lesson for the DeVries problem: compactification naturally gives more than one integer or representation label, and different sectors can couple to different gauge fields.

Open strings add boundary data. Neumann and Dirichlet boundary conditions relate left- and right-moving oscillators differently, and D-branes localize open-string states on submanifolds [2]. Coincident branes generate nonabelian gauge fields from open-string excitations. A DeVries derivation based on endpoints or branes must identify the two channels in Eq. (46) with actual boundary degrees of freedom, endpoint charges, or brane/bulk mixing modes.

The older phenomenological string lineage is relevant here. Regge trajectories in hadron spectra motivate a rotating flux-tube picture, and the open-string Veneziano amplitude was first written to capture strong-interaction features before QCD supplied the gauge-theory description [2]. The DeVries project uses this lineage as a methodological cue: a low-energy spectral regularity can be a real guide to a deeper string or boundary structure, provided the proposed determinant is eventually derived.

Endpoint physics gives the simplest analogy for charged vector states. In a coincident-brane system, strings stretching between branes carry off-diagonal gauge quantum numbers, and the massless excitations organize into a nonabelian gauge field [2]. The electroweak target is a broken nonabelian gauge sector coupled to a doublet order parameter. A successful endpoint model must therefore produce both the adjoint-current datum $J = 2$ and the doublet/order-parameter datum $J = 3/4$ inside one boundary or brane determinant.

A Kaluza–Klein or brane derivation would have to produce the determinant

$$\det \begin{pmatrix} x & -\sqrt{J} \\ -\sqrt{J} & x + J \end{pmatrix} = 0. \quad (46)$$

This can arise from a two-channel boundary problem, a mixing of brane-localized and bulk modes, or an effective orbit problem. The required output is the exact J -dependence in Eq. (46).

Extra-dimensional electroweak models show that boundary conditions can break gauge symmetry and determine vector spectra. Higgsless constructions on intervals provide a controlled language for relating boundary data, vector masses, fermion localization, and precision observables [7]. The interval viewpoint is valuable because boundary conditions can be obtained as limits of boundary dynamics. For the present project, this gives a precise target: derive the $J = 3/4, 2$ samples as boundary data associated with the Higgs doublet/order-parameter channel and the adjoint/current channel.

Witten’s search for realistic Kaluza–Klein theory gives the historical benchmark. Extra dimensions can generate gauge groups from geometry, including $SU(3) \times SU(2) \times U(1)$, while the realistic fermion quantum numbers impose severe constraints [12]. The DeVries program inherits this lesson. A compactification mechanism must solve the electroweak spectral problem and remain compatible with chiral matter.

M-theory on G_2 -holonomy manifolds supplies a separate completion map. Smooth compact G_2 compactifications give four-dimensional minimal supersymmetry in the large smooth limit. Realistic charged chiral matter requires singularities supporting nonabelian gauge sectors and chiral fermions [13–15]. Local Higgs-bundle descriptions over associative three-cycles give a modern way to organize gauge sectors, localized matter, and singular transitions [16].

The completion problem can therefore be stated as a list of mechanisms and obligations.

Mechanism	What it can supply	Required DeVries test
Regge/string tower	high-spin asymptotics	preserve slope and produce Eq. (46)
Circle KK reduction	gauge fields and charged KK modes	identify the charge/Casimir source of J
Winding sector	second compactification charge	test a winding-like interpretation of the second branch
Interval/brane BC	vector spectra and boundary dynamics	derive $J = 3/4, 2$ samples from boundary data
Endpoint/open-string sector	localized gauge states	identify the two channels as endpoint or brane/bulk modes
Singular G_2	gauge/chiral sectors	localize the electroweak operator geometrically
Higgs bundle	matter/mass localization	identify the two branches as spectral-cover data

Section status. This section states completion mechanisms and tests. A dynamical derivation requires a concrete boundary, compactification, or endpoint model producing Eq. (46).

VII. RADICAL PLACEMENT AND SCALE ATTACHMENT

The secular equation contains the radical

$$\Delta_J^{1/2} = \sqrt{J^2 + 4J}. \quad (47)$$

The manuscript treats $\Delta_J^{1/2}$ as the central object whose physical address must be identified. The algebra alone specifies a two-branch spectrum. Physics begins when the radical is assigned to a pole condition, a boundary determinant, a compactification eigenvalue, a worldsheet constraint, or a running matching condition.

A. Placement options

The possible placements are distinct research programs.

Placement	Physical meaning	Derivation owed
Pole spectrum	$\Delta_J^{1/2}$ enters a propagator or self-energy pole condition	derive the W/Z pole ratio and scalar part-ner branch from one spectral equation
GUT boundary	$\Delta_J^{1/2}$ fixes a high-scale weak-angle datum	embed the J -samples in a unification representation and run to the pole scale
Running parameter	$\Delta_J^{1/2}$ constrains a scheme-dependent coupling or vacuum parameter	specify the scheme, scale, matching map, and threshold corrections
KK eigenvalue	$\Delta_J^{1/2}$ comes from a compact internal wave operator	identify the internal operator, mode labels, degeneracies, and boundary conditions
Open-string end-point	$\Delta_J^{1/2}$ comes from or Chan–Paton data	derive the two-channel matrix from endpoint charges and brane/bulk mixing
Worldsheet strain	con- $\Delta_J^{1/2}$ comes from matching, conformal weights, or a modular sector	identify the CFT labels and show how they reduce to the electroweak determinant

The pole-spectrum placement is the current working reading. It gives the numerical clue its electroweak content because W and Z are observed as poles. It also gives the negative branch a task: the same determinant should know about the scalar/order-parameter side. This reading demands the most direct dynamical calculation, since it asks string/Kaluza–Klein structure to survive down to a low-energy pole observable.

The GUT-boundary placement gives a familiar comparison class. In SU(5) and Spin(10) settings, representation theory can give a clean weak-angle value at the high scale, with running and thresholds supplying the low-energy relation [20]. A DeVries high-scale reading would join that lineage. The current manuscript pursues the pole reading because the negative branch and pole-ratio clue appear together in the same low-energy normalization.

B. Scale attachment

Let $\mathcal{O}_J$ denote the physical operator whose two-channel reduction gives

$$\det(\lambda - \mathcal{O}_J) = \lambda^2 + J\lambda - J. \quad (48)$$

The scale question is then a question about where $\mathcal{O}_J$ lives:

$$\mathcal{O}_J \in \{\mathcal{O}_{\text{pole}}, \mathcal{O}_{\text{GUT}}, \mathcal{O}_{\text{KK}}, \mathcal{O}_{\text{brane}}, \mathcal{O}_{\text{CFT}}\}. \quad (49)$$

Each address has a different interpretation of the same algebra. A pole operator must be built from four-dimensional gauge-Higgs correlators. A GUT operator must be built from representation traces and matching. A KK operator must be a compact internal differential operator or boundary value problem. A brane operator must couple localized and bulk modes. A worldsheet operator must arise from conformal weights, current algebra, or level matching.

This list is useful because it prevents the radical from floating between schemes. The manuscript can change placement; each placement changes the required derivation.

C. Placement ledger

The placement question can be recorded as a ledger. A source theory $\mathcal{S}$ supplies an operator, the operator supplies a two-channel kernel, the kernel supplies branches, and the branches acquire physical meaning after a scheme and an assignment rule have been fixed:

$$\begin{aligned} \mathcal{S} &\longrightarrow \mathcal{O}_J \longrightarrow \mathcal{M}_J \longrightarrow \{x_+(J), x_-(J)\}, \\ \{x_+(J), x_-(J)\} &\longrightarrow \{s_{\text{pole}}^2, \mathcal{F}_{\text{sc}}, \mathcal{C}_{\text{top}}\}. \end{aligned} \quad (50)$$

Here $\mathcal{F}_{\text{sc}}$ denotes the scalar/order-parameter functional sought in Sec. V, and $\mathcal{C}_{\text{top}}$ denotes the global-form and representation constraints discussed in Secs. XIV and XV. The ledger separates four tasks:

$$\text{operator origin: } \mathcal{S} \mapsto \mathcal{O}_J, \quad (51)$$

$$\text{two-channel reduction: } \mathcal{O}_J \mapsto \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}, \quad (52)$$

$$\text{branch reading: } \mathcal{M}_J \mapsto \{x_+(J), x_-(J)\}, \quad (53)$$

$$\text{physical assignment: } \{x_{\pm}\} \mapsto (s_{\text{pole}}^2, \mathcal{F}_{\text{sc}}, \mathcal{C}_{\text{top}}). \quad (54)$$

Each arrow has different source support. Complex-pole and scheme language belongs to the last arrow [10]. Higgs doublet, adjoint current, custodial structure, and the electroweak ray belong to the assignment arrow [8, 9]. Endpoint labels, D-brane gauge fields, and interval boundary conditions belong to the first two arrows [2, 3, 7]. The current manuscript has exact control of Eq. (53); the other arrows are open derivation targets.

This ledger also organizes the negative branch. If a candidate source theory derives $\mathcal{M}_J$, then $x_-(J)$ is carried by the same light-sector basis as $x_+(J)$. The scalar question becomes a question about the eigenvector map,

$$u_-(J) \in \mathcal{H}_J \quad \longmapsto \quad \mathcal{F}_{\text{sc}}, \quad (55)$$

where $\mathcal{H}_J$ is the two-dimensional light sector. A Higgs interpretation requires $\mathcal{F}_{\text{sc}}$ to be gauge invariant. A brane interpretation requires $\mathcal{F}_{\text{sc}}$ to be a modulus, separation, or localized boundary datum. A compactification interpretation requires $\mathcal{F}_{\text{sc}}$ to be a normalized internal eigenmode or deformation parameter.

The ledger gives a compact criterion for progress. A new idea improves the paper when it fills one of the arrows in Eq. (50); it distracts from the paper when it adds labels outside the ledger.

D. Original research route

The route developed here is:

$$\begin{aligned} \text{Regge/high-spin datum} &\longrightarrow \text{two-channel secular determinant} \\ &\longrightarrow \text{electroweak pole ratio and scalar partner branch.} \end{aligned} \quad (56)$$

The first arrow belongs to the string/dual-model tradition: spectra organize into towers with controlled asymptotics. The second arrow is the DeVries algebra. The third arrow is the open electroweak calculation.

A concrete model should therefore supply four ingredients:

1. an internal label whose quadratic invariant is J ;
2. a two-channel system whose off-diagonal mixing is fixed by $\sqrt{J}$;
3. a diagonal trace term giving the $-J$ entry;

4. a physical normalization that places the positive branch on the vector pole ratio and the negative branch on the scalar/order-parameter side.

The fourth ingredient is the new conceptual burden. It turns the algebraic clue into a physical program and fixes the relation between the pole reading and high-scale weak-angle boundary data.

Section status. This section states the radical-placement problem and the original re-search route. The physical operator $\mathcal{O}_J$ remains open.

VIII. DUAL-MODEL LINEAGE AND OPEN-STRING ENDPOINT DATA

The DeVries construction enters the manuscript through a spectral regularity. This is why the older dual-model lineage matters. The Veneziano amplitude was introduced to capture strong-interaction regularities before its string interpretation became clear; its open-string form later supplied a world-sheet explanation of the pole tower and the associated Regge behavior [2]. The lesson for the present work is methodological and technical. A compact algebraic regularity in particle data can guide a deeper spectral construction when the amplitude or boundary problem subsequently explains the regularity.

The relevant feature is the coexistence of a low-level spectrum and a high-spin asymptotic rule. In string amplitudes, pole residues organize the spins carried by the exchanged string states, and the tower structure follows from the quantized oscillators [2]. In the DeVries program, the secular equation is required to do something analogous at a much lower and more constrained level:

$$\text{high-spin/Regge regularity} \quad \rightsquigarrow \quad x^2 + Jx - J = 0 \quad \rightsquigarrow \quad \{x_+(J), x_-(J)\}. \quad (57)$$

The first arrow is historical motivation until a world-sheet or boundary calculation produces the second expression. The second arrow is algebraic and already fixed. The manuscript therefore treats the Rivero/DeVries observation as a clue that selects a target determinant.

Open strings provide a more concrete mechanism than closed-string Regge language alone. An open-string state carries oscillator data in the interior and boundary data at its endpoints. D-branes give a geometric home to the endpoints, and coincident branes give nonabelian gauge fields from open-string modes [2, 3]. This suggests a precise kind of model for Eq. (46):

one channel should be an adjoint or current channel associated with gauge fields, while the other should carry the order-parameter data that samples the doublet invariant.

For the DeVries determinant, the endpoint route can be written as a quadratic action for two boundary-coupled amplitudes,

$$S_J^{(2)} = \begin{pmatrix} \psi_1^\dagger & \psi_2^\dagger \end{pmatrix} \begin{pmatrix} \lambda & -\kappa_J \\ -\kappa_J & \lambda + \tau_J \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}, \quad (58)$$

with the DeVries target

$$\kappa_J^2 = J, \quad \tau_J = J. \quad (59)$$

Here ψ_1, ψ_2 are placeholders for actual endpoint, brane, or localized modes. The notation is intentionally sparse. The physics is carried by the derivation of κ_J and τ_J .

There are three natural endpoint readings.

1. *Chan-Paton/current reading.* The endpoint labels carry gauge indices. The diagonal entry τ_J is a trace or current-algebra invariant, and the off-diagonal entry κ_J is a group-theoretic transition matrix element.
2. *Brane/bulk reading.* One amplitude is localized on a brane or boundary, while the other propagates in the bulk. The determinant comes from integrating over the bulk mode and imposing the boundary matching condition.
3. *Order-parameter reading.* One channel is an adjoint-current datum and the other is a doublet/order-parameter datum. This is the route that can explain why the electroweak samples are $J = 2$ and $J = 3/4$.

Each reading must respect the same electroweak constraint. The pole ratio compares charged and neutral vector poles, while the $J = 3/4$ datum belongs naturally to the Higgs doublet side. A successful endpoint construction therefore has to turn an order-parameter doublet invariant into the charged-vector sample in the pole quotient. This is the ordered-sampling problem in boundary language.

The open-string route also clarifies the negative branch. The two roots are eigenvalues of one two-channel system. If the positive branch is associated with the vector pole quotient, the negative branch should correspond to the partner channel in the same boundary problem. In an endpoint construction this partner may be a scalar boundary modulus, a Higgs-like

order parameter, a brane separation mode, or an auxiliary channel eliminated in the low-energy action. The manuscript uses the negative branch as a demand on the model: the same determinant has to explain both eigenvalues.

This places the string interpretation in a concrete form. The target is an endpoint or brane calculation whose reduced determinant has the entries in Eq. (59), whose low-energy normalization gives the W/Z pole quotient, and whose partner eigenvalue lands in the scalar/order-parameter sector.

Section status. This section is source-supported for the dual-model, open-string, and brane background. The endpoint determinant is conjectural and states the calculation owed.

IX. KALUZA–KLEIN SPECTRA, BOUNDARY CONDITIONS, AND SECULAR DETERMINANTS

Kaluza–Klein theory supplies the second concrete language for the DeVries program. The original idea unifies gauge fields with geometry by expanding a higher-dimensional theory around a compact ground state. In Witten’s account, symmetries of the compact space appear as gauge symmetries in the lower-dimensional theory, and the physical spectrum is obtained from small oscillations around the compactification background [12]. The same account also records the hard model-building burden: obtaining the Standard Model gauge group, fermion quantum numbers, and scalar sector is a serious representation problem, even when the gauge group itself can be engineered.

The DeVries problem asks for a narrower and sharper object inside this tradition: a compact or interval derivation of the two-channel determinant

$$\det \begin{pmatrix} \lambda & -\sqrt{J} \\ -\sqrt{J} & \lambda + J \end{pmatrix} = \lambda^2 + J\lambda - J. \quad (60)$$

This determinant is the entire target of the KK route.

A. Circle compactification and two charge channels

String compactification on a circle gives a clean source-backed example of a two-channel spectral structure. Ordinary Kaluza–Klein momentum is quantized along the circle. Closed strings add winding around the same circle. Polchinski’s circle example shows that the

Kaluza–Klein vector couples to compact momentum, while the vector descending from the antisymmetric tensor couples to winding [3]. At the self-dual radius the spectrum has enhanced current algebra, and changing the radius is described by an operator built from the same currents.

This gives the manuscript a useful analogy:

$$\left(\begin{array}{c} \text{momentum-like channel} \\ \text{winding-like or boundary channel} \end{array} \right) \rightsquigarrow \{x_+(J), x_-(J)\}. \quad (61)$$

The analogy becomes a derivation after the two channels are reduced to Eq. (60). The self-dual-radius example also explains why a scalar modulus belongs in the discussion. The compact radius is a scalar field in the lower-dimensional theory, and gauge enhancement can occur at special values of that scalar. This is the compactification version of the manuscript’s positive/negative branch problem: a vector spectral statement and a scalar modulus can arise from the same compact sector.

B. Interval variational data

Extra-dimensional field theory on an interval gives a more direct determinant language. Csaki–Hubisz–Meade derive boundary conditions from the variational principle for fields on an interval and show how boundary mass terms and boundary kinetic terms modify those conditions [7]. A scalar example has the schematic action

$$S_{\text{int}} = \int d^4x \int_0^L dy \frac{1}{2} \left[\partial_M \Phi \partial^M \Phi - V(\Phi) \right] + S_0[\Phi] + S_L[\Phi], \quad (62)$$

with boundary functionals S_0 and S_L . The boundary variation supplies equations at $y = 0$ and $y = L$. Boundary mass terms give Robin-type conditions. Boundary kinetic terms can make the boundary equation depend on the four-dimensional eigenvalue. This is exactly the structural feature needed by a secular determinant.

For a mode $\Phi(x, y) = \phi_\lambda(x) f_\lambda(y)$, the interval problem has the abstract form

$$\mathcal{L}_J f_\lambda(y) = \lambda f_\lambda(y), \quad 0 < y < L, \quad (63)$$

$$\mathcal{B}_{0,J}(\lambda) f_\lambda(0) = 0, \quad \mathcal{B}_{L,J}(\lambda) f_\lambda(L) = 0. \quad (64)$$

The DeVries route asks for a two-dimensional light sector $\mathcal{H}_J$ extracted from this problem. A convenient basis is

$$\mathcal{H}_J = \text{span}\{h_J, a_J\}, \quad (65)$$

where h_J is a boundary order-parameter amplitude and a_J is an adjoint/current amplitude. The interval derivation then reduces the bulk-plus-boundary problem to an effective kernel

$$K_J^{\text{int}}(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J}^{\text{int}}(\lambda) & \Sigma_{ha,J}^{\text{int}}(\lambda) \\ \Sigma_{ah,J}^{\text{int}}(\lambda) & \lambda + \Sigma_{aa,J}^{\text{int}}(\lambda) \end{pmatrix}. \quad (66)$$

The DeVries target is the boundary limit

$$\Sigma_{hh,J}^{\text{int}} = 0, \quad \Sigma_{aa,J}^{\text{int}} = J, \quad \Sigma_{ha,J}^{\text{int}} \Sigma_{ah,J}^{\text{int}} = J. \quad (67)$$

This is the interval version of the endpoint target in Sec. XI.

C. Dictionary for the interval entries

The interval construction separates the entries of Eq. (60) into physically addressable data:

Entry	Interval origin	Derivation target
$\Sigma_{hh,J}^{\text{int}} = 0$	order-parameter boundary	reference fix the subtraction by a symmetry, a brane condition, or a normalization convention
$\Sigma_{aa,J}^{\text{int}} = J$	boundary inertia, adjoint current trace, or brane kinetic term	identify the quadratic invariant that appears in the boundary equation
$\Sigma_{ha,J}^{\text{int}} \Sigma_{ah,J}^{\text{int}} = J$	overlap between boundary and bulk profiles, brane-localized mixing, or endpoint-current coupling	derive the invariant from a normalized overlap or generator product
$(J_H, J_{\text{adj}}) = (3/4, 2)$	Higgs/order-parameter channel and adjoint/current channel	derive the ordered electroweak sampling rule from the coupled interval problem

The table is intentionally strict. A boundary mass term by itself can produce many secular equations. The DeVries claim requires the trace and determinant data in Eq. (67) and the electroweak ordering of J -samples.

The possible origins of J are correspondingly limited:

$$J \in \{C_2(R_{\text{int}}), C_2(R_{\text{bdry}}), h_{\text{CFT}}, q_{\text{KK}}^2, \langle f_{\text{bdry}}, f_{\text{bulk}} \rangle^2\}. \quad (68)$$

Here C_2 denotes a quadratic invariant, h_{CFT} a conformal weight or current-algebra datum, q_{KK} a compact charge, and the overlap term a normalized boundary/bulk integral. The final paper should name one of these entries and derive the equality to J .

D. Gauge fields, boundary breaking, and the electroweak ray

Gauge fields on an interval carry a stronger constraint than scalar toy models. Boundary conditions that break gauge symmetry have to arise from the action and preserve the consistency of the lower-dimensional gauge theory. Csaki–Hubisz–Meade use this language for electroweak symmetry breaking by boundary conditions and for Higgsless models [7]. Their framework is relevant here because it gives a controlled setting in which massive vector modes, a photon zero mode, custodial structure, and brane-localized terms can be discussed in one spectral problem.

The DeVries construction requires the boundary model to preserve the electroweak ray. The low-energy gauge-Higgs mass matrix has the form

$$\mathcal{M}_{\text{EW}}^2(v) = v^2 \mathcal{P}(g, g'), \quad (69)$$

where v is radial and $\mathcal{P}(g, g')$ is projective. The DeVries ratio can act on $\mathcal{P}(g, g')$, or on a boundary operator whose pole condition reduces to that same projective datum. Thus the interval model should yield a coupled charged/neutral construction:

$$\left[\begin{array}{c} SU(2)_L \text{ charged sector} \\ SU(2)_L \times U(1)_Y \text{ neutral sector} \\ \text{photon zero mode} \end{array} \right] \longrightarrow \frac{M_W^2}{M_Z^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})}. \quad (70)$$

This requirement places the W and Z readings inside one gauge-Higgs or boundary system. The photon must remain the unbroken mode, and the massive neutral pole must carry the adjoint/current sample.

The interval version of the ordered assignment theorem is:

$$\begin{aligned} &\text{bulk gauge fields} + \text{boundary terms} + \text{order-parameter channel} \\ &\implies K_J^{\text{int}}(\lambda) \implies (J_W, J_Z) = (3/4, 2). \end{aligned} \quad (71)$$

This theorem is parallel to the endpoint theorem in Sec. XI. The two routes can eventually compete: endpoint labels emphasize matrix algebra, while interval boundary conditions emphasize variational origin and pole equations.

E. Scalar partner from interval data

The negative branch also has a KK address. Witten's review notes that compactification naturally produces scalar degrees of freedom, including radius moduli and possible charged scalar modes tied to symmetry breaking questions [12]. Polchinski's circle compactification likewise identifies the compact radius with a lower-dimensional scalar and shows that changing the radius is tied to current algebra at enhanced-symmetry points [3]. In interval language, scalar data can appear as boundary values, Wilson-line-like data, brane-localized order parameters, or moduli of the compact geometry.

The scalar-branch target is therefore

$$u_-(J) \in \mathcal{H}_J \quad \longmapsto \quad \mathcal{F}_{\text{KK}} \in \{R(x), H^\dagger H, A_5, \text{brane scalar, localized modulus}\}. \quad (72)$$

Here $u_-(J)$ is the negative-branch eigenvector of the two-channel kernel. A final derivation should identify $\mathcal{F}_{\text{KK}}$, state its gauge transformation properties, and explain its scheme relation to the low-energy Higgs/order-parameter variables.

F. Hard test for the KK route

The KK route becomes a hard test once it is expressed as the following chain:

$$\begin{aligned} \text{compact or interval source} &\longrightarrow \text{boundary variational problem} \longrightarrow K_J^{\text{int}}(\lambda), \\ K_J^{\text{int}}(\lambda) &\longrightarrow \lambda^2 + J\lambda - J \longrightarrow \{s_{\text{pole}}^2, \mathcal{F}_{\text{KK}}\}. \end{aligned} \quad (73)$$

The first arrow is source-backed by the KK and interval literature. The second arrow is the DeVries determinant derivation. The last arrow is the pole-placement and scalar-branch assignment. This chain gives the manuscript a concrete way to compare compactification, endpoint, and G_2 routes.

Section status. The KK circle, interval, and boundary-condition background is source-supported. The reductions in Eqs. (67), (71), and (72) are theorem targets.

X. TOY ENDPOINT-BOUNDARY OPERATOR

The previous sections list possible mechanisms. This section turns those mechanisms into a minimal toy operator. The toy model is a target for derivation, suitable for referee pressure.

Let $h_J(x)$ denote a boundary-localized order-parameter amplitude and $a_J(x)$ denote an adjoint/current amplitude. The quadratic light-sector action is

$$S_J^{(2)} = \int d^4x \begin{pmatrix} h_J^\dagger & a_J^\dagger \end{pmatrix} \begin{pmatrix} \lambda & -\kappa_J \\ -\kappa_J & \lambda + \tau_J \end{pmatrix} \begin{pmatrix} h_J \\ a_J \end{pmatrix}. \quad (74)$$

Here λ is the spectral variable after normalization. The DeVries target is

$$\tau_J = J, \quad \kappa_J^2 = J, \quad (75)$$

which gives

$$\det \begin{pmatrix} \lambda & -\sqrt{J} \\ -\sqrt{J} & \lambda + J \end{pmatrix} = \lambda^2 + J\lambda - J. \quad (76)$$

The fields h_J and a_J are placeholders with fixed jobs. The h_J channel must carry the doublet/order-parameter invariant when $J = 3/4$. The a_J channel must carry the adjoint/current invariant when $J = 2$. The off-diagonal entry κ_J is the physical mixing amplitude between the two channels. The diagonal entry τ_J is the trace datum of the adjoint/current side.

A. Boundary origin

Interval field theory supplies a mechanism for such a matrix. Boundary mass and kinetic terms modify the variational boundary conditions, and brane-localized kinetic terms can make the boundary condition depend on the four-dimensional eigenvalue [7]. A schematic bulk-plus-boundary system is

$$S_{\text{bulk}} = \int d^4x \int_0^L dy \frac{1}{2} \left[(\partial_M A_J)^2 + (\partial_M H_J)^2 \right] + S_{\text{bdry}}[A_J, H_J]. \quad (77)$$

After solving the bulk equations and restricting to the two light boundary amplitudes, the effective inverse propagator has the form

$$K_J^{\text{eff}}(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J}(\lambda) & \Sigma_{ha,J}(\lambda) \\ \Sigma_{ah,J}(\lambda) & \lambda + \Sigma_{aa,J}(\lambda) \end{pmatrix}. \quad (78)$$

The DeVries calculation asks for a normalization and a boundary limit in which

$$\Sigma_{hh,J} = 0, \quad \Sigma_{aa,J} = J, \quad \Sigma_{ha,J}\Sigma_{ah,J} = J. \quad (79)$$

This formulation is useful because each matrix entry has a physical address. $\Sigma_{hh,J}$ tests whether the order-parameter channel can be used as the zero reference. $\Sigma_{aa,J}$ tests whether the adjoint/current channel supplies the trace J . $\Sigma_{ha,J}$ tests whether endpoint, brane, or overlap physics supplies $\sqrt{J}$.

B. Endpoint interpretation

In an open-string or D-brane reading, a_J is an endpoint current or Chan–Paton-like gauge excitation, while h_J is a boundary order-parameter mode. Coincident branes naturally give nonabelian gauge fields from endpoint sectors [2, 3]. The toy model demands a more specific result: the endpoint algebra must reduce to the entries in Eq. (75).

The corresponding route is

$$\begin{aligned} \text{endpoint labels} &\longrightarrow \kappa_J^2 = J \\ \text{adjoint/current trace} &\longrightarrow \tau_J = J \\ \text{boundary normalization} &\longrightarrow \Sigma_{hh,J} = 0. \end{aligned} \quad (80)$$

This table defines the calculation. It also gives a compact way to compare candidate completions.

C. Electroweak matching

The toy operator must finally match the electroweak projective data. The desired chain is

$$\begin{aligned} \text{boundary data} &\longrightarrow \mathcal{M}_J \longrightarrow \{x_+(J), x_-(J)\} \\ &\qquad\qquad\qquad \downarrow \end{aligned} \quad (81)$$

W/Z pole ratio and scalar branch.

The positive branch should yield

$$\frac{M_W^2}{M_Z^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})}, \quad (J_H, J_{\text{adj}}) = (3/4, 2). \quad (82)$$

The negative branch should identify the scalar or boundary partner in the same basis. The toy model therefore joins the three open calculations: ordered electroweak sampling, determinant derivation, and scalar-branch interpretation.

Section status. This section is a conjectural toy target. It translates the string/Kaluza–Klein mechanism into named matrix entries and exact derivation obligations.

XI. ENDPOINT GAUGE DATA AND BRANE HIGGSING

The toy operator in Sec. X becomes useful once its entries can be assigned to recognizable string or boundary data. Open strings and D-branes give one sharp language for this assignment. The route follows the original string-theory ambition: a particle spectrum is encoded in worldsheet and endpoint data, and spacetime fields arise as low-energy modes of that spectrum.

A. Endpoint labels as a source of matrix structure

An open string ending on a stack of N branes carries endpoint labels. In modern language these labels are Chan–Paton data. The corresponding low-energy fields sit in $N \times N$ matrices; for coincident branes, the massless excitations form a $U(N)$ gauge field and adjoint scalar fields on the brane worldvolume [2]. Polchinski’s review states the same logic in compactification language: current algebras of the compact worldsheet theory give space-time gauge bosons, and D-brane boundary conditions give open-string gauge fields along the brane [3].

For the present manuscript the important point is the matrix origin. Endpoint labels give a natural place for diagonal and off-diagonal channels:

$$|\psi; m, n\rangle, \quad m, n = 1, \dots, N, \quad (83)$$

with fields

$$A_a = (A_a)^m{}_n, \quad \Phi^I = (\Phi^I)^m{}_n. \quad (84)$$

The diagonal entries describe strings whose endpoints lie on the same brane. The off-diagonal entries describe strings stretched between distinct branes. At coincidence the off-diagonal vector modes complete the nonabelian gauge multiplet. This is precisely the type of two-channel bookkeeping needed by Eq. (11).

The source-backed part of the argument ends here. The DeVries-specific step is sharper

and remains open: the endpoint algebra must produce the two entries

$$\kappa_J^2 = J, \quad \tau_J = J, \quad (85)$$

with $J = 3/4$ and $J = 2$ entering in the ordered electroweak way. The endpoint labels supply a matrix arena; selecting Eq. (85) requires additional dynamics.

B. Worldvolume Yang–Mills and adjoint scalars

The low-energy action for coincident D-branes contains the nonabelian Yang–Mills field strength and adjoint scalars [2]. Schematically,

$$S_{\text{brane}} = \int d^{p+1}\xi \, \text{Tr} \left[-\frac{1}{4g_{\text{YM}}^2} F_{ab} F^{ab} - \frac{1}{2} D_a \Phi^I D^a \Phi^I + \frac{1}{4} [\Phi^I, \Phi^J]^2 + \dots \right]. \quad (86)$$

The schematic form is enough for the present program. Gauge fields and scalars occupy the same matrix algebra. The covariant derivative

$$D_a \Phi^I = \partial_a \Phi^I + i[A_a, \Phi^I] \quad (87)$$

couples off-diagonal vector modes to scalar expectation values. When two branes are separated, the off-diagonal gauge fields acquire a mass proportional to the separation; equivalently, the mass is the energy of a string stretched between the branes [2]. The brane construction therefore gives a geometric Higgs mechanism with a scalar modulus and charged vector modes in one matrix system.

This structure is the reason the endpoint route is attractive for the DeVries problem. The positive branch is meant to describe a vector pole-ratio clue. The negative branch asks for a scalar or order-parameter partner. A brane worldvolume action naturally contains both vector and scalar matrix entries. The current draft asks whether a two-dimensional reduction of such entries can be normalized to

$$\mathcal{M}_J = \begin{pmatrix} 0 & \kappa_J \\ \kappa_J & -\tau_J \end{pmatrix}, \quad \kappa_J^2 = \tau_J = J. \quad (88)$$

C. Endpoint dictionary for the DeVries entries

The endpoint hypothesis can now be stated as a dictionary. Let h_J be the boundary order-parameter amplitude and a_J the endpoint-current amplitude. Let $\mathcal{H}_J = \text{span}\{h_J, a_J\}$

be the two-dimensional light sector. A candidate derivation must provide three pieces of data:

$$\text{zero reference: } \langle h_J | K_J | h_J \rangle = 0, \quad (89)$$

$$\text{trace entry: } \langle a_J | K_J | a_J \rangle = J, \quad (90)$$

$$\text{mixing entry: } \langle h_J | K_J | a_J \rangle \langle a_J | K_J | h_J \rangle = J. \quad (91)$$

Here K_J is the normalized quadratic kernel obtained after integrating out heavier world-volume or bulk modes. Eq. (90) should be traced to a current algebra, adjoint Casimir, or boundary kinetic term. Eq. (91) should be traced to an endpoint overlap, a brane-localized coupling, or a generator normalization. Eq. (89) fixes the baseline for the order-parameter channel.

The representation data suggest the following table:

DeVries entry	Endpoint or brane origin	Derivation owed
$\tau_J = J$	adjoint/current trace, bound- ary kinetic term, or compact current algebra	show the normalized trace is the relevant quadratic invari- ant
$\kappa_J^2 = J$	endpoint overlap, stretched- string mixing, brane/order-parameter coupling	show the off-diagonal product or equals the same invariant
$\Sigma_{hh,J} = 0$	order-parameter channel	reference justify the subtraction by a symmetry, boundary condi- tion, or renormalization con- vention
$(J_H, J_{\text{adj}}) = (3/4, 2)$	doublet order parameter and adjoint current	derive the ordered W/Z sam- pling rule

This dictionary is restrictive. A completion must determine the entries, the basis, and the normalization from the source theory.

D. Scalar branch from brane separation

The brane picture also gives a natural language for the negative branch. In the D-brane Higgs mechanism, scalar expectation values describe brane positions. Off-diagonal vector fields become massive when the branes separate. Near coincidence, the off-diagonal modes become light and the full matrix structure controls the local physics [2]. Thus a two-branch matrix can plausibly pair a vector branch with a scalar or separation branch.

For the DeVries construction, this becomes the following target. There should be a scalar or boundary modulus φ_J and a normalized kernel K_J such that

$$K_J|_{\mathcal{H}_J} \sim \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}, \quad \varphi_J \text{ is read from } u_-(J). \quad (92)$$

Here $u_-(J)$ is the eigenvector of the negative eigenvalue. The phrase “read from” is an obligation: it must become a gauge-invariant map from the eigenvector data to a scalar quantity such as $H^\dagger H$, a brane separation, a boundary modulus, or a Higgs-potential parameter.

The scalar branch should also respect the electroweak ray. If the compact or brane model is meant to reproduce the Standard Model Higgs mechanism, the radial vacuum parameter v controls the common mass scale while the DeVries quotient controls the projective direction. A brane-separation interpretation must therefore identify which modulus supplies the radial scale and which normalized matrix supplies the angular data.

E. Why the endpoint route is a hard test

The endpoint route makes the proposal more constrained. It asks for an actual reduction:

$$\begin{aligned} \text{endpoint data} &\longrightarrow \text{worldvolume kernel} \longrightarrow \mathcal{M}_J, \\ \mathcal{M}_J &\longrightarrow \{x_+(J), x_-(J)\} \longrightarrow \text{pole ratio and scalar branch.} \end{aligned} \quad (93)$$

Each arrow has a check. The first arrow is source-supported by Chan–Paton labels, D-brane gauge fields, and adjoint scalars. The second arrow is the DeVries-specific determinant derivation. The third arrow is the exact algebra of Sec. II. The last arrow is the pole-placement and negative-branch problem.

This is also where the seventies lineage becomes substantive. The original dual amplitude began as a spectral object for hadron physics, and open-string amplitudes expose a tower of

particle poles [2]. The present manuscript uses that lineage as method: start from a spectral clue, search for the worldsheet or boundary object that produces it, then translate the result into the electroweak pole scheme. The general string-theory idea that particle data can arise from endpoint and compactification spectra is source-backed; the electroweak translation is speculative.

Section status. The endpoint, brane gauge-field, and brane Higgsing background is source-supported. The identification of the endpoint kernel with Eq. (88) is conjectural and supplies a concrete derivation target.

XII. G_2 LOCALIZATION AND LOCAL SPECTRAL OPERATORS

The G_2 route supplies a third concrete language for the DeVries program. Its role in this manuscript is local. It gives a framework in which gauge fields, charged matter, order-parameter data, and deformation modes can live at singular loci of a compactification. M-theory on G_2 -holonomy spaces is a natural route to four-dimensional $\mathcal{N} = 1$ structure; singularities provide the gauge and matter sectors needed for particle physics [13–15]. The DeVries question is whether a localized G_2 operator can reduce to the same two-channel determinant used in the pole-ratio clue.

A. Singular support for gauge and matter sectors

The source-backed starting point is the localization of degrees of freedom. Witten explains that G_2 compactifications with singularities can support abelian or nonabelian gauge groups with chiral fermions, with anomaly inflow from the bulk supplying a consistency condition [13]. Acharya and Witten describe examples where singular loci carry nonabelian gauge groups and conical or intersection points support chiral multiplets [14]. Acharya and Gukov review the broader picture: singular G_2 spaces are the essential setting for nonabelian gauge groups, chiral fermions, and strongly coupled gauge dynamics in M-theory compactifications [15].

The DeVries problem asks for a local sector with three named pieces:

$$\left(\begin{array}{c} \text{singular gauge locus} \\ \text{localized charged matter or order parameter} \\ \text{deformation mode of the singularity} \end{array} \right) \rightsquigarrow \mathcal{M}_J. \quad (94)$$

The singular gauge locus is the natural address for adjoint/current data. The localized charged matter or order-parameter mode is the natural address for the doublet side. The deformation mode is the natural address for the scalar branch.

This local focus also controls the topology issue. A compact G_2 model imposes global constraints, including anomaly, representation, and line-operator data. The determinant can be tested locally first. A global compactification can then be asked to support the same local operator and the Standard Model global form.

B. Higgs-bundle variables on an associative three-cycle

Modern local G_2 gauge sectors can be organized by a Higgs-bundle-like system on an associative three-cycle M_3 . Braun, Cizel, Hubner, and Schafer-Nameki describe a gauge field together with a one-form Higgs field on M_3 , with matter localized at critical loci and interactions controlled by gradient-flow data [16]. In the abelianized local description, the Higgs field can be written schematically as

$$\Phi = \sum_{i=1}^n t_i df_i, \quad \rho = \sum_{i=1}^n t_i \rho_i, \quad \Delta f_i = \rho_i, \quad (95)$$

where t_i are Cartan generators and the ρ_i encode charge distributions. For a representation with charge vector $Q = (q_1, \dots, q_n)$, the effective Morse function is

$$f_Q = \sum_{i=1}^n q_i f_i, \quad \rho_Q = \sum_{i=1}^n q_i \rho_i. \quad (96)$$

Matter in the representation R_Q localizes at points where

$$df_Q(p) = 0. \quad (97)$$

Flow lines between such critical points can generate mass terms, and trivalent flow trees can generate Yukawa couplings when the representation charges sum to zero [16]. This is

a strong local dictionary: gauge data, matter localization, mass terms, and interactions are all functions of the same geometric input.

For the DeVries program, the Higgs-bundle variables give a possible address for the local operator:

$$\mathcal{O}_J^{G_2} = \mathcal{O}_J(W, \Phi, \rho_Q, f_Q), \quad (98)$$

where W is the gauge field on M_3 . The precise form of $\mathcal{O}_J^{G_2}$ is open. The useful point is that W , Φ , and f_Q already separate adjoint gauge data, scalar deformation data, and localized representation data.

C. Local two-channel kernel

Let a_J be an adjoint/current fluctuation supported along the singular gauge locus, and let h_J be a localized order-parameter fluctuation at a critical point of f_Q . Let

$$\mathcal{H}_J^{G_2} = \text{span}\{h_J, a_J\}. \quad (99)$$

After integrating out heavier localized and bulk modes, the local inverse propagator or spectral kernel has the same abstract form as the endpoint and interval kernels:

$$K_J^{G_2}(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J}^{G_2}(\lambda) & \Sigma_{ha,J}^{G_2}(\lambda) \\ \Sigma_{ah,J}^{G_2}(\lambda) & \lambda + \Sigma_{aa,J}^{G_2}(\lambda) \end{pmatrix}. \quad (100)$$

The DeVries target is

$$\Sigma_{hh,J}^{G_2} = 0, \quad \Sigma_{aa,J}^{G_2} = J, \quad \Sigma_{ha,J}^{G_2} \Sigma_{ah,J}^{G_2} = J. \quad (101)$$

This target is a local statement. It can be tested near one singular locus or one conical point before a compact global model is specified.

D. Dictionary for the G_2 entries

The local G_2 dictionary is:

DeVries entry	G_2 origin	Derivation target
$\Sigma_{aa,J}^{G_2} = J$	ADE singular gauge sector, current algebra, or adjoint fluctuation along M_3	show that the normalized local trace is the relevant quadratic invariant
$\Sigma_{ha,J}^{G_2} \Sigma_{ah,J}^{G_2} = J$	localized matter overlap, line coupling, bifundamental vev, or singularity-deformation mixing	derive the off-diagonal product from the local wavefunctions or flow data
$\Sigma_{hh,J}^{G_2} = 0$	reference order-parameter channel at a critical point of f_Q	justify the zero entry through local normalization or symmetry of the critical-point problem
$(J_H, J_{\text{adj}}) = (3/4, 2)$	localized doublet/order-parameter data and joint/current data	derive the ordered electroweak assignment inside the same local singularity model

This table turns G_2 geometry into a determinant problem. It also prevents the geometry from becoming a decorative completion. Each geometric object must provide one entry of $K_J^{G_2}$, one assignment rule, or one consistency check.

E. Higgsing as singular deformation

Acharya and Witten give a pattern directly relevant to this manuscript: in examples where singular loci meet, gauge groups live on the singular components, chiral multiplets can be supported at the meeting point, and Higgsing corresponds to a geometric deformation or smoothing of the singularity [14]. In their brane-related examples, a vev for a bifundamental field can break a product gauge symmetry to a diagonal subgroup, while the local singular geometry changes accordingly.

For DeVries, this suggests the deformation chain

$$\begin{aligned}
&\text{intersecting singular loci} \longrightarrow \text{localized chiral/order-parameter mode} \\
&\longrightarrow \text{deformed singularity} \longrightarrow K_J^{G_2}(\lambda).
\end{aligned} \tag{102}$$

The scalar branch would then be read from the deformation direction. If $u_-(J)$ is the negative-branch eigenvector in $\mathcal{H}_J^{G_2}$, the target scalar map is

$$u_-(J) \mapsto \mathcal{F}_{G_2},$$

$$\mathcal{F}_{G_2} \in \left\{ \begin{array}{l} \text{deformation parameter, } f_Q(p), \rho_Q, \\ \text{bifundamental vev, localized modulus} \end{array} \right\}. \quad (103)$$

The map must also respect the electroweak ray. The deformation parameter can supply the scalar side, while the pole-ratio branch supplies the projective vector side.

F. Anomaly and global checks

The G_2 route has strong consistency tests. Witten's anomaly-inflow discussion links local singularity data to four-dimensional chiral matter and anomaly cancellation [13]. Therefore a local DeVries operator has to pass the following checks:

1. the singular gauge locus supports the required adjoint sector;
2. localized charged matter has the desired representation content;
3. anomaly inflow or its local analogue is compatible with the charged fields;
4. the global gauge-group form of Sec. XV is compatible with the local representations;
5. flavor labels remain separate from compactification topology until a representation map is derived.

These checks can reject a candidate completion even when the two-channel determinant is obtained locally.

G. Hard test for the G_2 route

The G_2 route becomes a hard test in the same form as the endpoint and interval routes:

$$\begin{aligned} \text{singular } G_2 \text{ neighborhood} &\longrightarrow (W, \Phi, \rho_Q, f_Q) \longrightarrow K_J^{G_2}(\lambda), \\ K_J^{G_2}(\lambda) &\longrightarrow \lambda^2 + J\lambda - J \longrightarrow \{s_{\text{pole}}^2, \mathcal{F}_{G_2}, \mathcal{C}_{\text{top}}\}. \end{aligned} \quad (104)$$

The first arrow is source-backed by singular G_2 compactification and Higgs-bundle literature. The second arrow is the DeVries determinant derivation. The final arrow is the pole-placement, scalar-branch, and global-form assignment. A future proof can start locally, then extend to compact global data after the determinant appears.

Section status. The G_2 localization, anomaly, Higgsing, and Higgs-bundle background is source-supported. The reductions in Eqs. (101), (103), and (104) are theorem targets.

XIII. COMPARISON OF THE THREE LOCAL ROUTES

The endpoint, interval, and G_2 sections introduce three ways to place the DeVries determinant inside a theory with localized gauge and order-parameter data. This section puts them in a common format. The purpose is to make the next analytical work precise: each route must deliver the same normalized two-channel kernel, the same electroweak assignment, the same branch interpretation, and compatible global data. The comparison also prevents conceptual drift. A successful route supplies a physical origin for entries of a matrix; it gives more than an analogy with strings, extra dimensions, or singular geometry.

A. Common reduction problem

Let $\mathcal{S}_r$ denote one of the three route descriptions,

$$r \in \{\text{end}, \text{int}, G_2\}, \quad \mathcal{S}_r = \begin{cases} \mathcal{S}_{\text{end}} & \text{open-string endpoint or D-brane sector,} \\ \mathcal{S}_{\text{int}} & \text{interval Kaluza-Klein boundary sector,} \\ \mathcal{S}_{G_2} & \text{localized singularity sector.} \end{cases} \quad (105)$$

The desired reduction chooses a two-dimensional light subspace

$$\mathcal{H}_{J,r} = \text{span}\{h_{J,r}, a_{J,r}\}, \quad (106)$$

where $h_{J,r}$ is the order-parameter channel and $a_{J,r}$ is the adjoint/current channel. In this basis the reduced spectral kernel has the form

$$K_{J,r}(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J}^r(\lambda) & \Sigma_{ha,J}^r(\lambda) \\ \Sigma_{ah,J}^r(\lambda) & \lambda + \Sigma_{aa,J}^r(\lambda) \end{pmatrix}. \quad (107)$$

The route passes the first DeVries test when a stated limit and normalization give

$$\Sigma_{hh,J}^r = 0, \quad \Sigma_{aa,J}^r = J, \quad \Sigma_{ha,J}^r \Sigma_{ah,J}^r = J. \quad (108)$$

Equation (108) is stronger than a qualitative mechanism. It assigns one geometric, endpoint, or boundary object to each matrix entry. It also fixes the algebraic route to

$$\det K_{J,r}(\lambda) = \lambda^2 + J\lambda - J, \quad \lambda \in \{x_+(J), x_-(J)\}. \quad (109)$$

The reduction has three addresses. The endpoint address is supported by the open-string fact that endpoint labels and coincident branes generate matrix-valued gauge fields, with brane positions supplying scalar data [2, 3]. The interval address is supported by extra-dimensional models in which boundary conditions and brane-localized terms set vector spectra [7]. The G_2 address is supported by singular compactifications that localize gauge sectors and chiral matter, together with Higgs-bundle variables organizing local matter and interaction data [13–16]. These sources justify the addresses. They leave Eq. (108) as the shared derivation target.

B. Entry-by-entry comparison

The first comparison concerns the diagonal current entry. In the DeVries matrix the current channel carries the trace contribution. The route-specific question is therefore:

$$a_{J,r} \rightsquigarrow \Sigma_{aa,J}^r = J. \quad (110)$$

For endpoints, $a_{J,\text{end}}$ should be a Chan–Paton or brane gauge amplitude. The current trace would then come from a normalized endpoint algebra or an adjoint current on a coincident-brane stack. For intervals, $a_{J,\text{int}}$ should be the boundary value of a bulk gauge mode or a current coupled to it; the trace contribution would come from the boundary kinetic or mass term after solving the bulk equation. For G_2 , a_{J,G_2} should be a fluctuation along a singular gauge locus, with J arising from an ADE or adjoint invariant in the local gauge sector.

The second comparison concerns the off-diagonal product. The determinant requires a product of transition amplitudes,

$$h_{J,r} \leftrightarrow a_{J,r}, \quad \Sigma_{ha,J}^r \Sigma_{ah,J}^r = J. \quad (111)$$

For endpoints, this product should arise from an open string stretched between sectors, a brane-Higgsing field, or an endpoint overlap. For intervals, it should arise from boundary matching between a localized order parameter and a bulk gauge profile. For G_2 , it should arise from a localized matter wavefunction, flow-line overlap, bifundamental vev, or deformation coupling at a singular point. The route earns physical content when the product is computed from normalized wavefunctions, endpoint algebra, or local geometry.

The third comparison concerns the order-parameter reference entry,

$$h_{J,r} \rightsquigarrow \Sigma_{hh,J}^r = 0. \quad (112)$$

The zero entry is a normalization and symmetry statement. In an endpoint model, it may come from choosing the boundary order-parameter amplitude as the reference channel. In an interval model, it may come from a boundary condition or brane term that subtracts the local order-parameter self-energy at the pole. In a G_2 model, it may come from the critical-point normalization of the localized mode. A derivation must name the symmetry, subtraction, or normalization that gives this reference value.

The following table records the three entries in a format useful for a referee.

Kernel entry	Endpoint/D-brane route	Interval KK route	G_2 route
$\Sigma_{aa} = J$	adjoint Chan–Paton boundary current or coincident-brane gauge amplitude	value of a ADE bulk gauge mode with fluctuation or kinetic data	singular gauge invariant
$\Sigma_{ha}\Sigma_{ah} = J$	stretched string, end-point overlap, or brane-Higgsing field	localized parameter to a bulk profile	order-matter wavefunction coupling overlap, flow line, bifundamental vev, or singular deformation
$\Sigma_{hh} = 0$	boundary channel for the order parameter	reference normalized order-order-parameter energy	boundary critical-point self-ization for localized matter

C. Electroweak assignment as a route-independent target

The same electroweak assignment target applies to every route. The low-energy field-theory data contain an $SU(2)_L$ Higgs doublet and adjoint gauge currents. Their natural quadratic invariants give

$$J_H = \frac{3}{4}, \quad J_{\text{adj}} = 2. \quad (113)$$

The target map is

$$\mathcal{A}_r : (\mathcal{S}_r, \mathcal{H}_{J,r}) \longrightarrow (J_W, J_Z) = (J_H, J_{\text{adj}}). \quad (114)$$

This map is a theorem target because the observed W and Z vector bosons are both gauge-sector particles after symmetry breaking. The route must explain why the charged comparison samples the order-parameter invariant while the neutral comparison samples the adjoint/current invariant. A useful route therefore produces the ordered pair from the gauge-Higgs coupling structure, from endpoint boundary labels attached to the Higgsing field, from interval boundary data that distinguish charged and neutral sectors, or from a localized singularity model where doublet and adjoint data meet.

The pole-ratio reading then has the common form

$$\frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})} \quad (115)$$

after the pole scheme of Sec. III has been fixed. This equation states the desired low-energy placement. A route that naturally yields the same quotient at a compactification scale or unification scale would still be useful, because it would locate the algebra in a source theory. The pole reading would then require an additional matching theorem.

D. Branch interpretation and scalar data

Every route diagonalizes the same two-channel kernel. Therefore each route has to account for both eigenvectors. Let

$$u_{\pm,J,r} = \alpha_{\pm,J,r} h_{J,r} + \beta_{\pm,J,r} a_{J,r} \quad (116)$$

be normalized eigenvectors of the route kernel in the DeVries limit. The positive branch supplies the pole-ratio clue. The negative branch supplies the scalar-side question,

$$u_{-,J,r} \rightsquigarrow \mathcal{F}_{\text{sc},r}. \quad (117)$$

The functional $\mathcal{F}_{\text{sc},r}$ depends on the route. In the endpoint route it may be a brane separation, a stretched-string scalar, or a brane-Higgsing vev. In the interval route it may be a boundary order-parameter functional, a radion-like deformation, or a localized scalar eigenmode. In the G_2 route it may be a singularity-deformation parameter, a critical value of f_Q , a charge-density functional ρ_Q , or a localized modulus.

The scalar map has to be tied to the electroweak ray. The gauge-Higgs system has a radial vacuum scale and a projective vector-mass ratio. A route that derives the determinant should identify how the positive branch fixes the projective vector datum and how the negative branch enters the radial or scalar sector. This is the point where the two-root structure becomes physically restrictive. A determinant used for the pole clue automatically supplies its partner branch, and the route must assign that partner branch a controlled role.

E. Scale-placement ledger

The route comparison separates three scale placements:

$$\text{pole scale,} \quad \text{compactification or boundary scale,} \quad \text{unification scale.} \quad (118)$$

The manuscript presently takes the pole scale as the proposed attachment point for the electroweak clue. This choice gives the project its strongest phenomenological motivation. The GUT lineage supplies the comparison class: a clean weak-angle value can be a high-scale structural datum, then acquire low-energy meaning through running and thresholds. In the present construction the clean value is attached to the pole ratio, so the scale-placement theorem is logically prior to any compactification geometry.

For endpoint and interval routes, the most direct challenge is the relation between a boundary determinant and a four-dimensional pole. Boundary conditions usually define a spectrum before loop corrections and before the conversion to pole observables. The route must explain how the determinant survives matching to the pole definition, or how the pole reading emerges from a dressed boundary self-energy.

For the G_2 route, the challenge is sharper. A local singularity model naturally lives at a compactification or high scale. The local kernel can organize representations and couplings, yet a pole-spectrum reading requires a low-energy matching chain. The useful outcome is

therefore a two-stage theorem:

$$\mathcal{S}_{G_2} \longrightarrow K_{J,G_2} \longrightarrow \Delta_{V,T}^{-1}(s) \quad \text{with} \quad s = s_{W,Z}. \quad (119)$$

The first arrow is local geometry. The second arrow is four-dimensional matching. The current draft states both as open obligations.

F. Referee decision tree

The comparison can be used as a decision tree for the next research pass:

1. Choose one source theory $\mathcal{S}_r$ and one light subspace $\mathcal{H}_{J,r}$.
2. Derive the reduced kernel $K_{J,r}(\lambda)$ from an action, boundary condition, endpoint algebra, or local G_2 operator.
3. Identify the normalization giving Eq. (108).
4. Derive the ordered electroweak map in Eq. (114).
5. Match the positive branch to the pole observable or state the matching theorem required for that placement.
6. Map the negative branch to a scalar, modulus, brane, or deformation functional.
7. Check global-form, anomaly, and representation compatibility.

Each step has a defined failure mode. Extra light channels enlarge the determinant. A different trace or mixing product gives a different secular relation. A high-scale placement changes the interpretation of the weak-angle clue. An incompatible global form rejects a proposed compactification. These outcomes would still be useful research results because they would classify the DeVries relation inside a larger family of spectral constructions.

Section status. This section is open-program material. It compares three source-backed local routes and states the common reduction, assignment, branch, scale-placement, and global-compatibility tests.

XIV. FLAVOR BOUNDARY AND REPRESENTATION ALGEBRA

The electroweak mass-ratio problem and the flavor problem carry separate obligations in this manuscript. The working boundary condition is:

$$\begin{aligned} &\text{DeVries electroweak determinant} \\ &\text{and generation/flavor organization} \\ &\text{are linked only by an explicit model.} \end{aligned} \tag{120}$$

This prevents a compactification or endpoint picture from turning a useful flavor pattern into an electroweak derivation by naming alone.

Representation theory remains essential. In the Standard Model and in the familiar grand-unification lineage, fermions live in representations of the internal gauge group, while gauge bosons live in the adjoint sector [19, 20]. $SU(5)$, $Spin(10)$, Pati–Salam, and E_6 show how a larger algebra can unify the Standard Model representation data. This is the correct role of the GUT material in the present paper: it supplies representation discipline and a comparison class for structural weak-angle statements.

The DeVries construction uses a different representation question. The electroweak determinant asks why one sample is associated with the Higgs/order-parameter doublet invariant and the other with the adjoint/current invariant:

$$J_H = T_H(T_H + 1) = 3/4, \quad J_{\text{adj}} = T_{\text{adj}}(T_{\text{adj}} + 1) = 2. \tag{121}$$

Flavor representations can constrain a completion. The missing step is a derived map from endpoint, compactification, or localized matter data into the two-channel electroweak operator.

The $SO(32)$ -flavor idea should therefore be treated as an organizing boundary sector. In an endpoint language it may label external or internal flavor degrees of freedom, Chan–Paton-like data, or a bookkeeping algebra for fermion multiplicities. Its required checks are concrete:

1. identify the representation space carrying the $SO(32)$ -flavor labels;
2. map the Standard Model chiral fermions into that space;
3. show anomaly cancellation and hypercharge normalization;

4. state the relation to the global quotient $\Gamma \subset \mathbb{Z}_6$;
5. couple the flavor sector to the electroweak determinant only through a derived operator.

This keeps the flavor idea useful while protecting the central claim. The electroweak ratio concerns W/Z pole structure and the Higgs/order-parameter channel. Generation structure concerns chiral matter, Yukawa textures, and possible endpoint or compactification multiplicities. A string/Kaluza–Klein completion may eventually bind them together. The binding has to be an equation or a local geometric construction.

The compactification version has an additional caveat. Geometry can generate gauge symmetries, chiral localization, and multiple wavefunctions with the same gauge quantum numbers [12, 14, 16]. That makes topology relevant to flavor. The manuscript still treats generation counting as a separate task because the DeVries determinant is a two-channel electroweak object. The decisive test is whether the same local model that yields generations also yields

$$\det(\lambda - \mathcal{O}_J) = \lambda^2 + J\lambda - J \quad (122)$$

with the ordered electroweak samples in Eq. (121).

The practical rule for later sections is therefore simple. Flavor material may enter as a constraint on endpoints, compactification, anomaly cancellation, and global form. It may motivate a local operator. The W/Z mass-ratio argument rests on the electroweak determinant and its pole placement.

Section status. This section states a source-supported boundary condition. The SO(32)-flavor interpretation remains conjectural until a representation map and coupling to $\mathcal{O}_J$ are supplied.

XV. GLOBAL FORM OF THE STANDARD MODEL GAUGE GROUP

The local Standard Model gauge algebra admits several global forms. The gauge group may be written as

$$G_{\text{SM}} = \frac{SU(3)_C \times SU(2)_W \times U(1)_Y}{\Gamma}, \quad \Gamma \subset \mathbb{Z}_6, \quad (123)$$

with $\Gamma = 1, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_6$ as the standard possibilities [17, 18]. Local perturbative fields see the Lie algebra. Line operators, magnetic sectors, theta-angle periodicities, and topological probes see the quotient.

This is relevant to the DeVries program for two reasons. First, any compactification, endpoint, or G_2 completion has to specify the global group it realizes. Second, a pole-spectrum determinant for the electroweak sector has to coexist with the same line-operator spectrum as the low-energy Standard Model. Thus a proposed $\mathcal{O}_J$ comes with a global-form compatibility condition:

$$\mathcal{O}_J \text{ is defined in a theory with } G_{\text{SM}} = (SU(3) \times SU(2) \times U(1))/\Gamma. \quad (124)$$

Tong’s line-operator analysis classifies how the quotient changes the electric, magnetic, and dyonic line sectors [17]. The quotient restricts admissible electric line operators and changes the allowed magnetic line operators through Dirac quantization. It also changes theta-angle periodicities. After electroweak symmetry breaking, the line sectors descend to electromagnetic probes, so Γ leaves a low-energy topological signature.

Hsin and collaborators propose a transport-style probe of this global form through a fractional topological response coefficient [18]. The important point for this manuscript is structural: the global form is, in principle, observable through nonlocal or topological response data. It supplies a completion constraint for the DeVries program, because a string or compactification model generally determines more than a local Lie algebra.

Grand unification provides a useful comparison. $SU(5)$ and $\text{Spin}(10)$ naturally package Standard Model representations in a way that favors the $\mathbb{Z}_6$ quotient, and the subgroup $S(U(2) \times U(3))$ realizes the quotient structure inside $SU(5)$ [19, 20]. This is another reason the GUT lineage is the correct conceptual comparison class for a structural electroweak angle. The DeVries pole reading must still specify why its clean value belongs to low-energy poles; the GUT lineage shows what high-scale structural placement looks like.

The global-form constraint can be expressed as a commutative diagram requirement. If a compactification or endpoint model gives an ultraviolet group G_{UV} , then the low-energy electroweak sector should fit into

$$\begin{array}{ccc} G_{\text{UV}} & \longrightarrow & G_{\text{SM}} \\ \downarrow & & \downarrow \\ \mathcal{O}_{\text{UV}} & \longrightarrow & \mathcal{O}_J, \end{array} \quad (125)$$

where the upper arrow fixes Γ and the lower arrow reduces the ultraviolet spectral operator to the DeVries two-channel operator. A model that supplies $\mathcal{O}_J$ while leaving Γ unspecified has an incomplete topological sector.

The mass-ratio argument itself should use local electroweak fields and pole positions. The global-form material acts as a filter on completions. A viable string/Kaluza–Klein/ G_2 derivation should state:

1. which quotient Γ appears;
2. which line operators survive compactification and electroweak breaking;
3. how the Higgs/order-parameter field transforms under the chosen global group;
4. how the DeVries operator respects the same quotient.

Section status. This section is source-supported and functions as a completion constraint. It supplies topological consistency conditions for future compactification or endpoint derivations.

XVI. STATUS OF THE ANALYTICAL PROGRAM AND CONCEPTUAL TESTS

The construction yields a precise number from a precise branch quotient. The electroweak comparison becomes meaningful after choosing the pole-mass observable and stating the placement convention. The numerical fact has clue status. The project advances when the following calculations are completed.

1. Derive the ordered assignment $(J_W, J_Z) = (3/4, 2)$ from the gauge-Higgs sector or from a boundary/compactification problem.
2. Derive the two-channel determinant Eq. (46) from a dynamical model.
3. Determine whether the negative branch maps to the Higgs instability or to another auxiliary sector.
4. Update the pole-mass comparison with current source-audited inputs and uncertainty propagation.
5. Specify the global form of the gauge group in any proposed topological completion.

The conceptual tests are correspondingly direct. A source-audited pole-mass ratio far from Eq. (9) would reduce the value of the motivating clue. A gauge-Higgs or boundary construction that yields Eq. (29) would turn the electroweak assignment into a derivation. A compactification model that produces Eq. (46) and the negative-branch scalar scale would give the string/Kaluza–Klein interpretation physical content. A dynamical derivation that produces a different ordered pair or a different branch quotient would redirect the project toward a broader class of secular mass relations.

A. Claim hierarchy

The manuscript uses four claim levels. This hierarchy is part of the physics content because it tells a reader which statements are established and which statements are theorem targets.

Level	Manuscript content	Required evidence
Derived	algebra of $Q(J)$, roots, branch identities, trace and determinant form	direct equation-level derivation in Sec. II and App. C
Source-supported	electroweak mass matrix, cited local source fragments and pole/running scheme distinctions, endpoint/brane, KK, and G_2 background	explicit references
Conjectural	pole placement of the DeVries quotient, negative-branch scalar reading, route-specific kernel entries	theorem targets with assumptions and failure outcomes
Open	ordered J -assignment, full determinant derivation, scalar functional, boundary condition, endpoint algebra-global-form compatibility of a compact completion	future proof from a stated action, boundary condition, endpoint algebra, or localized operator

This hierarchy controls how the result should be read. The exact result is the existence of a two-root secular object whose positive-branch quotient lies near the pole weak-angle clue and whose negative branch points toward scalar/order-parameter data under the project

normalization. The physical claim begins after a route derives the determinant and the ordered assignment.

B. Minimum viable derivation

A minimal derivation has five components:

$$\mathcal{S} \longrightarrow \mathcal{H}_J \longrightarrow K_J(\lambda) \longrightarrow (J_H, J_{\text{adj}}) \longrightarrow (s_{\text{pole}}^2, \mathcal{F}_{\text{sc}}). \quad (126)$$

The source theory $\mathcal{S}$ may be a boundary effective action, an endpoint sector, or a localized G_2 system. The light subspace $\mathcal{H}_J$ must contain an order-parameter channel and an adjoint/current channel. The kernel $K_J(\lambda)$ must reduce to the DeVries determinant in a stated normalization. The ordered pair $(J_H, J_{\text{adj}}) = (3/4, 2)$ must arise from gauge-Higgs or source-theory data. The final arrow must attach the positive branch to the pole observable and the negative branch to a defined scalar functional.

The derivation therefore has a compact acceptance criterion:

$$\det K_J(\lambda) = \lambda^2 + J\lambda - J, \quad (J_W, J_Z) = (3/4, 2), \quad u_-(J) \mapsto \mathcal{F}_{\text{sc}}. \quad (127)$$

This criterion is deliberately demanding. A partial result can still improve the manuscript by proving one arrow in Eq. (126). For example, an interval model may derive the determinant class while leaving the pole placement open; an endpoint model may derive the ordered representation pair while leaving the scalar map open; a G_2 model may supply a strong localization mechanism while leaving four-dimensional pole matching open.

C. Decision ledger for route selection

The three routes should be compared by the same criteria:

Criterion	Endpoint/D-brane	Interval KK	G_2 localization
Trace source	endpoint current or ad-joint brane gauge field	boundary value of bulk gauge mode	a singular gauge locus or ADE invariant
Mixing source	stretched string or brane-Higgsing mode	boundary overlap of scalar and gauge pro-files	flow-line, matter wave-function, or deformation coupling
Scalar branch	brane separation or stretched-string scalar	boundary scalar, like mode, radion, or brane order parameter	A_5 -singularity deformation, critical value, charge density, or localized modulus
Scale challenge	endpoint determinant to four-dimensional pole	boundary spectrum to dressed pole	local compactification data to low-energy pole
Global check	gauge representation and endpoint labels	chiral matter and gauge group from compact matter, global form space	anomaly inflow, local

The most useful next calculation is the one that fixes a matrix entry from source data. The order of attack should favor the entry with the clearest address. Endpoint physics gives a clear address for the off-diagonal mixing through stretched strings and brane-Higgsing fields. Interval physics gives a clear address for the kernel through boundary variation. G_2 localization gives a clear address for representation data and scalar deformation variables. A research pass that proves any one of these entries should immediately update the theorem-target appendix.

D. Referee-facing tests

The paper can be evaluated by concrete tests:

1. The pole-scheme test: every W/Z comparison must use pole masses or give a conversion to pole quantities.
2. The assignment test: $J = 3/4$ must come from a doublet/order-parameter datum and

$J = 2$ from an adjoint/current datum.

3. The determinant test: the source theory must produce the trace and determinant data of Eq. (46).
4. The scalar test: the negative branch must map to a defined scalar, boundary, brane, or deformation functional.
5. The ray test: the construction must respect the full electroweak broken-to-unbroken ray.
6. The global test: a compact completion must be compatible with Standard Model representations, line operators, and anomaly constraints.

These tests are severe enough to make the speculative proposal useful. They allow a route to be rejected cleanly. They also allow a route to generate a broader secular family when the DeVries entries arise as a limit.

E. Current manuscript status

The current draft has converted the original observation into a source-controlled research program. The algebraic object is explicit. The pole observable is defined. The electroweak assignment is stated as a theorem target. The negative branch has a scalar-functional target. The string, endpoint, interval, and G_2 routes have been placed into a common kernel language. Flavor and global-form material have been separated from the determinant derivation and treated as compatibility checks.

The remaining burden is analytical. The strongest next result would be a derivation of Eq. (108) from one route. The next strongest result would be a derivation of the ordered electroweak assignment from gauge-Higgs data. A scalar-branch derivation would also be central because it would explain why Hans de Vries's positive-branch clue was accompanied by a hidden scalar-side clue in the same quadratic.

Section status. This section states the program's current logical dependencies, acceptance criteria, and route-selection tests.

Appendix A: Algebraic details

For completeness, we record the elementary algebra. From Eq. (5),

$$x_{\pm}(J) = \frac{-J \pm \sqrt{J(J+4)}}{2}. \quad (\text{A1})$$

The positive branch has the large- J expansion

$$x_+(J) = 1 - \frac{1}{J} + \frac{2}{J^2} + O(J^{-3}), \quad (\text{A2})$$

while

$$x_-(J) = -J - 1 + \frac{1}{J} + O(J^{-2}). \quad (\text{A3})$$

Thus the positive branch approaches a fixed asymptotic value while the negative branch carries the large negative partner required by the trace.

For $J = 3/4$,

$$x_+\left(\frac{3}{4}\right) = \frac{-3/4 + \sqrt{57}/4}{2} = \frac{\sqrt{57} - 3}{8}. \quad (\text{A4})$$

For $J = 2$,

$$x_+(2) = \frac{-2 + \sqrt{12}}{2} = \sqrt{3} - 1. \quad (\text{A5})$$

Therefore

$$s_{\text{dV}}^2 = 1 - \frac{\sqrt{57} - 3}{8(\sqrt{3} - 1)}. \quad (\text{A6})$$

Appendix B: Numerical reproducibility

The script ‘calculations/devries_spectrum.py’ computes the branch roots and the derived mass value from a supplied M_Z . The central functions are

```
x_plus(J) = (sqrt(J*J + 4*J) - J)/2
sin2_devries = 1 - x_plus(3/4)/x_plus(2)
MW = MZ * sqrt(1 - sin2_devries)
```

The test file ‘calculations/tests/test_devries_spectrum.py’ checks the root identities, the numerical value of s_{dV}^2 , and the seed value obtained from $M_Z = 91.1880 \text{ GeV}$.

Appendix C: Theorem targets for a derivation

This appendix records the analytical obligations in theorem form. The purpose is operational. Each target names its hypotheses, conclusion, current evidence, and failure mode. A complete version of the manuscript should either prove these statements, replace them with stronger statements, or demote the dependent claims.

1. Common format

The targets share a common data structure. Let $\mathcal{S}$ be a source theory or effective description. It may be a four-dimensional electroweak theory, a boundary effective action from an interval compactification, an open-string endpoint sector, or a localized G_2 system. Let $\mathcal{E}_J$ be the two-dimensional light subspace that is meant to carry the order-parameter and current amplitudes. The desired reduction has the form

$$\mathcal{S} \longrightarrow \mathcal{E}_J \longrightarrow \mathcal{M}_J = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix} \longrightarrow \{x_+(J), x_-(J)\}. \quad (\text{C1})$$

The reduction is meaningful after the normalization of $\mathcal{E}_J$ and the scheme in which the resulting eigenvalues are read have been specified. The pole-observable section fixes the electroweak comparison as a complex-pole statement, with Breit-Wigner/on-shell and running quantities treated as translated inputs [10, 11]. The electroweak section fixes the low-energy field-theory data that any assignment theorem must respect [5, 8, 9]. The string and Kaluza–Klein sections list candidate sources of the two-channel determinant [2, 3, 7].

For each target below, “proof” means an explicit derivation from a stated action, boundary condition, operator algebra, or compactification mode equation. A plausibility argument may motivate the target, yet the manuscript’s final physical claim must track the proven statement.

2. Target I: pole placement

Hypotheses. Assume a source theory $\mathcal{S}$ gives a two-branch spectral equation with roots $x_{\pm}(J)$. Assume the electroweak comparison is made with transverse gauge-boson complex

poles,

$$\det \Delta_{V,T}^{-1}(s_V) = 0, \quad s_V = M_{V,\text{pole}}^2 - iM_{V,\text{pole}}\Gamma_{V,\text{pole}}. \quad (\text{C2})$$

Assume common experimental mass conventions are converted into the pole convention before forming

$$s_{\text{pole}}^2 = 1 - \frac{M_{W,\text{pole}}^2}{M_{Z,\text{pole}}^2}. \quad (\text{C3})$$

Conclusion. The DeVries branch quotient is a pole-spectrum datum:

$$s_{\text{pole}}^2 = 1 - \frac{x_+(J_H)}{x_+(J_{\text{adj}})} \quad (\text{C4})$$

up to stated radiative, threshold, and scheme translations. The theorem must also explain why the assigned scale is the low-energy pole spectrum, with high-scale boundary placements handled by a separate derivation.

Current evidence. The manuscript already separates complex poles, Breit-Wigner/on-shell masses, $\overline{\text{MS}}$ running angles, effective weak angles, GUT boundary values, and compactification eigenvalues. Martin–Robertson supply the scheme language used in Sec. III. The CDF-II result has descriptive displacement status in the current draft.

Failure mode. The target fails if the derived clean value lives at a high-scale boundary, a running $\overline{\text{MS}}$ angle, an effective leptonic angle, or a compactification eigenvalue with a different translation law. In that case the pole-ratio reading becomes a motivating clue, and the manuscript must relocate the physical statement to the derived scheme.

3. Target II: electroweak assignment

Hypotheses. Let H be the Standard Model Higgs doublet and let $J_H = 3/4$ be its $SU(2)_L$ quadratic invariant. Let $J_{\text{adj}} = 2$ be the corresponding adjoint invariant for $SU(2)_L$ currents. The gauge-Higgs sector contains the usual mass terms

$$M_W^2 = \frac{g^2 v^2}{4}, \quad M_Z^2 = \frac{(g^2 + g'^2)v^2}{4}, \quad M_\gamma^2 = 0, \quad (\text{C5})$$

with the custodial structure and $\rho = 1$ tree-level relation discussed in Sec. IV.

Conclusion. There exists a natural map from gauge-Higgs data to ordered samples,

$$\mathcal{D}_{\text{EW}} : (H, J_\mu^a, B_\mu) \mapsto (J_W, J_Z) = (J_H, J_{\text{adj}}) = (3/4, 2), \quad (\text{C6})$$

such that

$$\frac{M_W^2}{M_Z^2} = \frac{x_+(J_H)}{x_+(J_{\text{adj}})}. \quad (\text{C7})$$

The derivation must say why the charged comparison samples the order-parameter invariant while the neutral comparison samples the adjoint/current invariant.

Current evidence. The Higgs doublet and adjoint invariants are fixed representation-theoretic facts. The electroweak mass matrix, photon null direction, custodial relation, and projective reading of $g : g'$ are standard low-energy data [5, 8, 9]. The manuscript has converted the assignment into a theorem target.

Failure mode. The target fails if the assignment uses observed vector spin or an arbitrary post-selection of J values. A successful derivation must arise from the gauge-Higgs variables, a boundary operator coupled to them, or a compactification mechanism whose low-energy reduction yields the ordered pair.

4. Target III: two-channel boundary determinant

Hypotheses. Let h_J be an order-parameter amplitude and a_J an adjoint/current amplitude in a light boundary subspace. Let their normalized quadratic kernel be

$$K_J(\lambda) = \begin{pmatrix} \lambda + \Sigma_{hh,J} & \Sigma_{ha,J} \\ \Sigma_{ah,J} & \lambda + \Sigma_{aa,J} \end{pmatrix}. \quad (\text{C8})$$

The source may be an interval theory with brane-localized terms, an open-string endpoint sector, or a localized singularity construction. Boundary mass and kinetic terms can alter eigenvalue conditions in extra-dimensional models [7]. Open-string endpoints and D-brane gauge sectors supply a separate endpoint language [2, 3].

Conclusion. There is a controlled limit and normalization in which

$$\Sigma_{hh,J} = 0, \quad \Sigma_{aa,J} = J, \quad \Sigma_{ha,J}\Sigma_{ah,J} = J, \quad (\text{C9})$$

so that

$$\det K_J(\lambda) = \lambda^2 + J\lambda - J. \quad (\text{C10})$$

Equivalently, the physical derivation fixes the two named entries

$$\tau_J = J, \quad \kappa_J^2 = J. \quad (\text{C11})$$

Current evidence. Sec. X gives the target action and the required entries. The literature sources support the existence of boundary-condition and endpoint mechanisms, while the exact DeVries entries remain conjectural.

Failure mode. The target fails if boundary variation gives a different trace, a different off-diagonal product, or an additional channel that changes the determinant class. The useful result in that case would be a larger family of secular relations containing the DeVries matrix as a limiting or approximate case.

5. Target IV: negative branch as scalar data

Hypotheses. Assume Targets I–III identify a physical two-channel matrix whose eigenvalues are $x_{\pm}(J)$. Let $u_{\pm}(J)$ be normalized eigenvectors. The negative branch is then part of the same diagonalization as the positive branch:

$$\mathcal{M}_J u_{\pm}(J) = x_{\pm}(J) u_{\pm}(J). \quad (\text{C12})$$

The scalar side of the electroweak theory contains the gauge-invariant radial datum, the Higgs pole, and the parameters of the symmetry-breaking potential [8, 9].

Conclusion. There exists a gauge-invariant scalar functional $\mathcal{F}_{\text{sc}}$ such that

$$\mathcal{F}_{\text{sc}}(H^{\dagger}H, D_{\mu}H, \text{boundary data}) = C |x_{-}(J_{\star})| \quad (\text{C13})$$

for a stated $J_{\star}$, normalization C , and scheme. Candidate readings include a Higgs-potential parameter, the radial vacuum scale, a boundary modulus, or a brane-separation mode.

Current evidence. Sec. V records the scalar target table and the eigenvector interpretation. The current draft treats the negative branch as an analytical partner of the pole clue, with its Higgs/order-parameter meaning open.

Failure mode. The target fails if the negative branch has purely auxiliary meaning in the derived source theory. The positive-branch clue can still motivate the determinant problem, while the scalar claims must then be removed from the physical conclusion.

6. Target V: global-form and localization compatibility

Hypotheses. Assume a string, brane, Kaluza–Klein, or G_2 source derives Targets II–IV. The low-energy gauge group has global form

$$G_{\text{SM}} = \frac{SU(3) \times SU(2) \times U(1)}{\Gamma}, \quad \Gamma \subseteq \mathbb{Z}_6. \quad (\text{C14})$$

Line operators and global identifications provide tests of the Standard Model gauge group beyond the Lie algebra [17, 18]. Singular G_2 spaces can localize gauge and chiral matter sectors in M-theory constructions [13–16].

Conclusion. The derived operator $\mathcal{M}_J$ is compatible with the allowed line operators, representation assignments, anomaly constraints, and localized matter data of the chosen completion. A successful completion must specify whether flavor labels are endpoint data, compactification data, representation-sector data, or a combination constrained by the source geometry.

Current evidence. Secs. XIV and XV state the current boundary conditions: the $\text{SO}(32)$ -flavor idea is a representation-sector clue, and global-form data are completion constraints. The manuscript avoids identifying generations with compactification topology until a representation map is derived.

Failure mode. The target fails if the determinant arises in a gauge or flavor sector incompatible with Standard Model line operators, anomaly conditions, or observed representations. A failure here would be decisive for a proposed completion even if the local two-channel determinant is mathematically correct.

7. Referee checklist

The theorem targets lead to the following checklist for future revisions:

1. State the source theory $\mathcal{S}$ and the light subspace $\mathcal{E}_J$.
2. Specify the normalization that makes $\Sigma_{hh,J} = 0$, $\Sigma_{aa,J} = J$, and $\Sigma_{ha,J}\Sigma_{ah,J} = J$.
3. Derive the ordered assignment $(J_H, J_{\text{adj}}) = (3/4, 2)$ from low-energy gauge-Higgs or boundary data.
4. Give the pole-scheme translation for every quoted W/Z mass input.

5. Identify the negative branch as scalar data, auxiliary data, or a derived sector with another physical role.
6. Check global-form, anomaly, and representation compatibility.

This checklist is the manuscript’s current definition of a complete derivation.

Section status. This appendix is open-program material. It states theorem targets and source-backed constraints; the proofs remain future analytical work.

Appendix D: Codex task map

The agentic workflow used for this manuscript is encoded in ‘codex_tasks/’. The tasks separate algebra, pole-mass scheme, electroweak interpretation, string/Kaluza–Klein/ G_2 completion, flavor boundary, global-form appendix, and referee review. This separation is useful because the numerical clue and the conceptual gap are different objects. The numerical clue is the combined positive- and negative-branch pattern. The conceptual gap is the derivation of Eq. (29) and the placement of the secular radical on the low-energy pole spectrum.

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